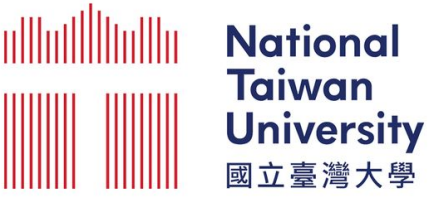
Text as Data

W9 · Word Embeddings and Political Language

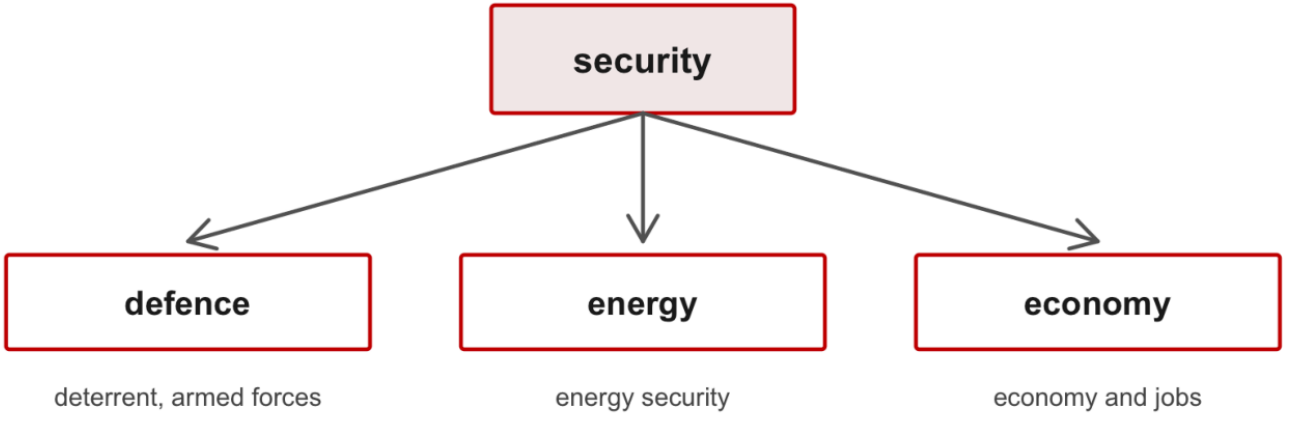
Yen-Chieh Liao

2026-11-06

Fall 2026 (115-1) · National Taiwan University (GIND)
NTLDEV 7191 · Fri. 16:30–18:20 · GIND 204



Example: security in context



Static embeddings blend these uses; contextual embeddings can separate them.

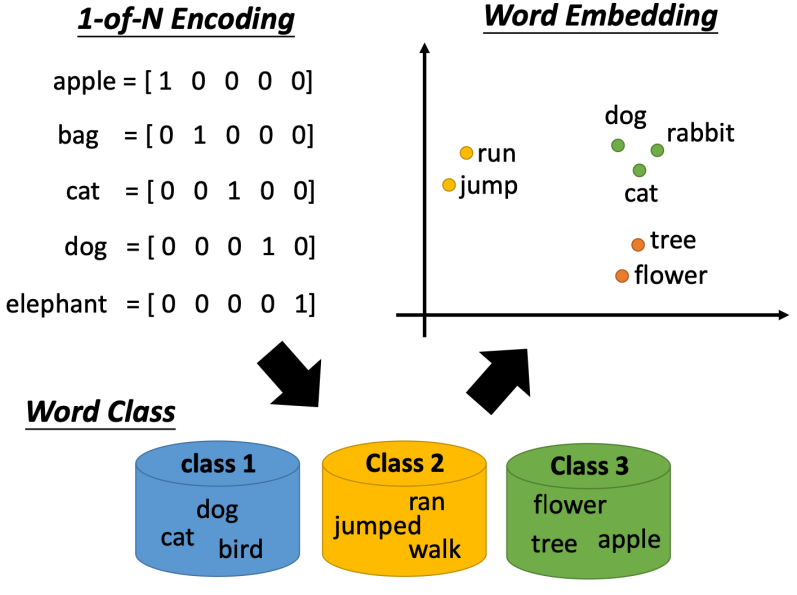
We see *security* used in defence, energy and economic contexts.

Which nearby words help us distinguish these uses?

How could we test whether these uses vary across parties or periods?

From topics to language representations

Research task	Political question
→ **W8 · Topic models**	What do actors discuss, and how much?
W9 · Word embeddings	How can learned vectors describe and compare political language?



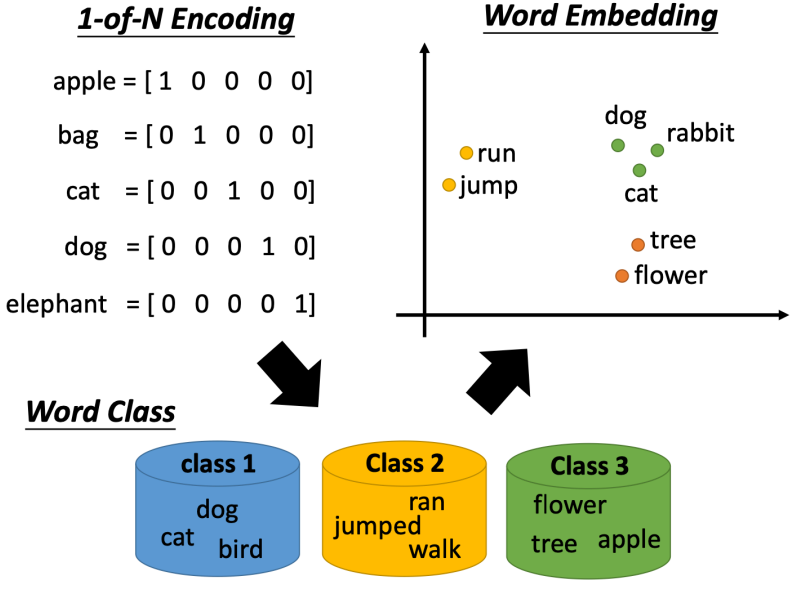
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Embeddings let us compare words using coordinates learned from their use.

Dog, *cat* and *rabbit* can be close when they appear in similar contexts.

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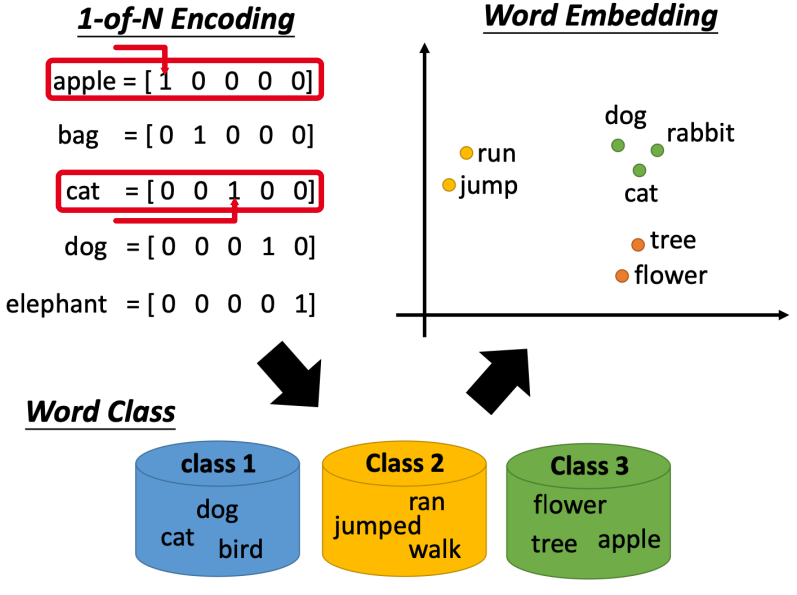
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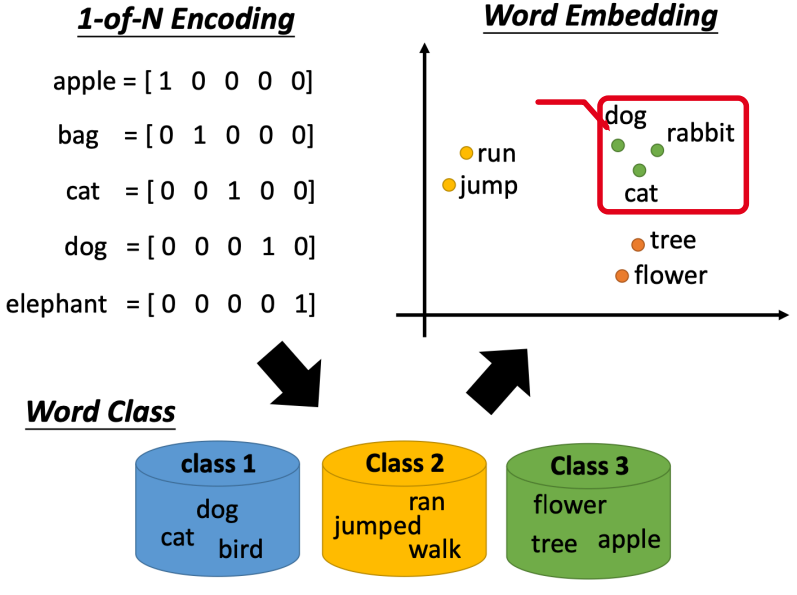
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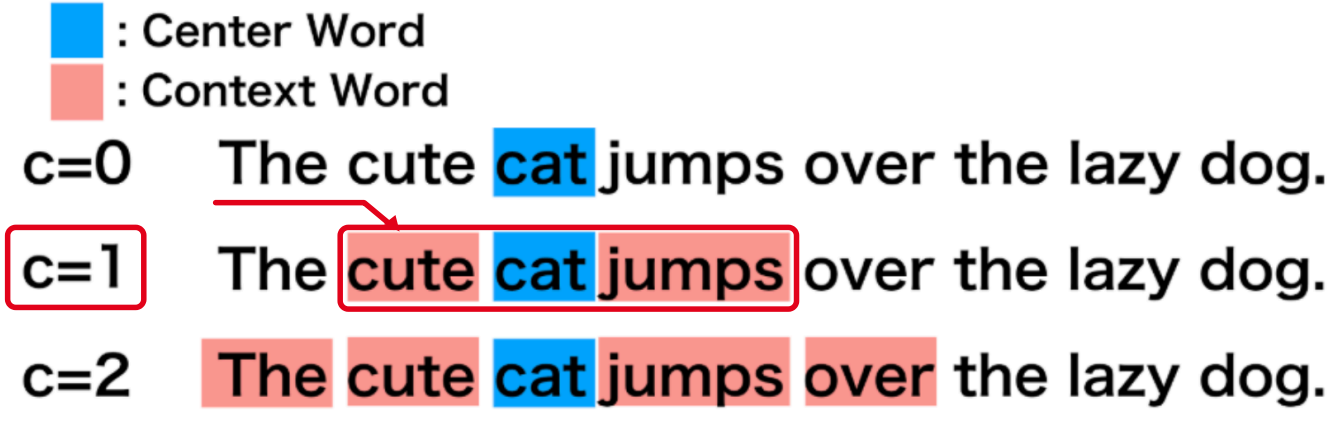


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Learn from context and compare vectors



Window · c = 1: We take *cute* and *jumps* around *cat*.

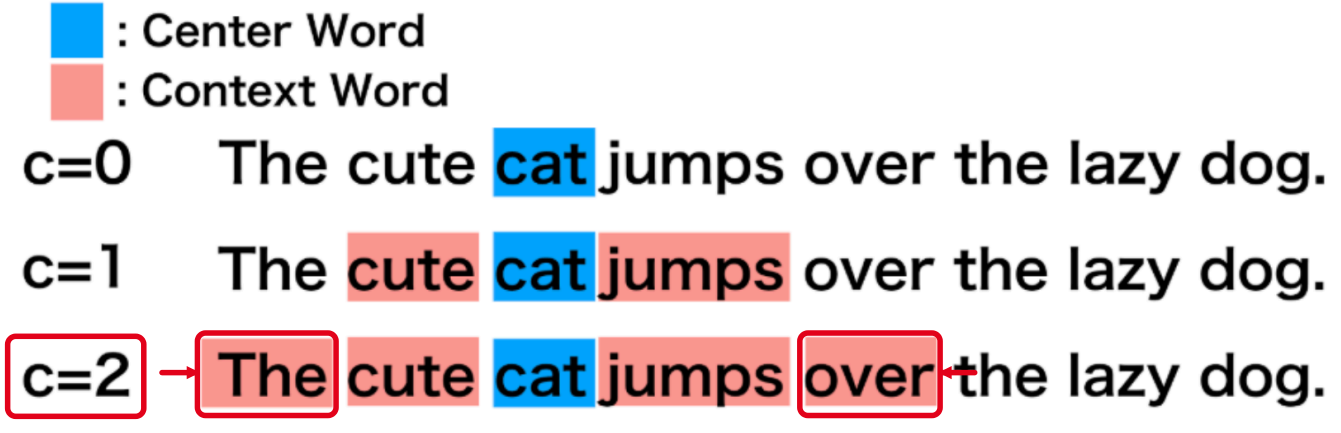
Window · c = 2: We also include *The* and *over*.

Learn: CBOW (**Continuous Bag of Words**) predicts *cat* from context. Skip-gram predicts context from *cat*.

Compare: We use cosine to compare the directions of the learned vectors.

Context window diagram · *Text as Data*, §8.2.4, p. 83 · Cosine, §7.1

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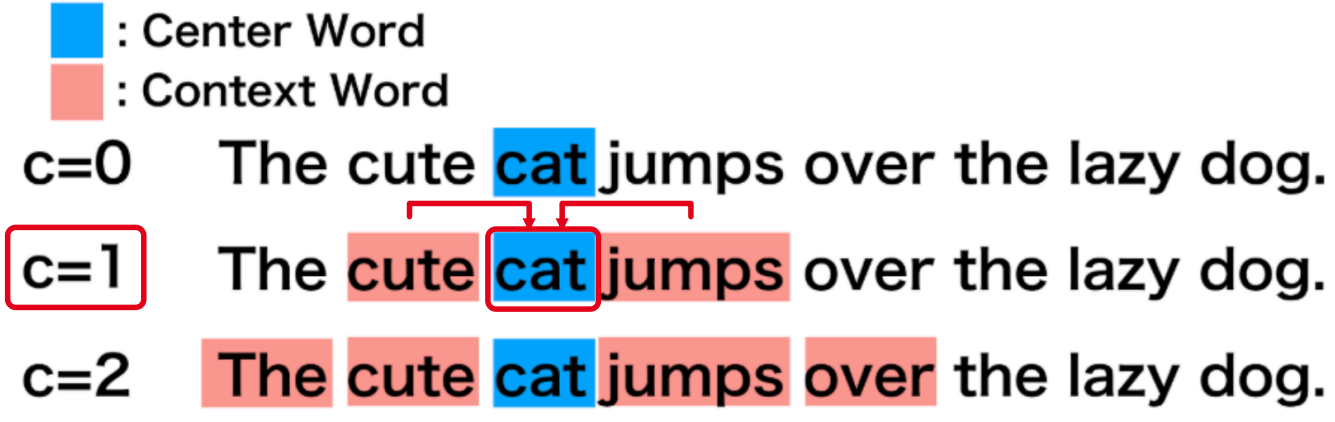
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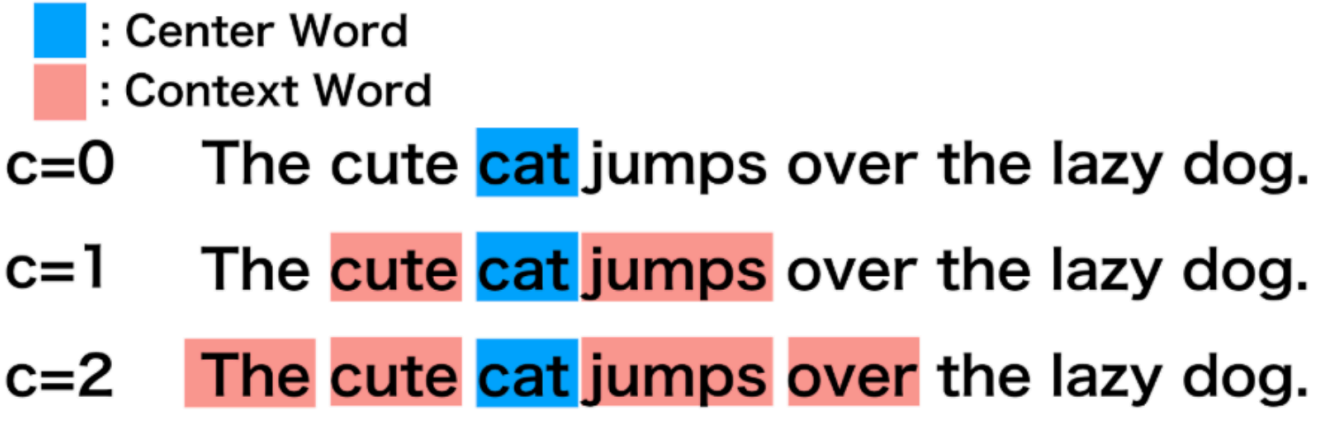
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Training choices shape the available vectors

Method or interface	Learning signal	Representation
→ **Word2Vec**	CBOW: context → word. Skip-gram: word → context.	Static word vector
GloVe Global Vectors for Word Representation	Fit corpus-wide word co-occurrence statistics.	Static word vector
fastText	Predict from context with character n-grams.	Static word vector
FlairEmbeddings	Character language models use surrounding text.	Contextual token vector

Lab choice: We use Flair's `WordEmbeddings("glove")` to load **static GloVe**.

Text as Data, pp. 83–84 · GloVe · fastText · Flair contextual embeddings

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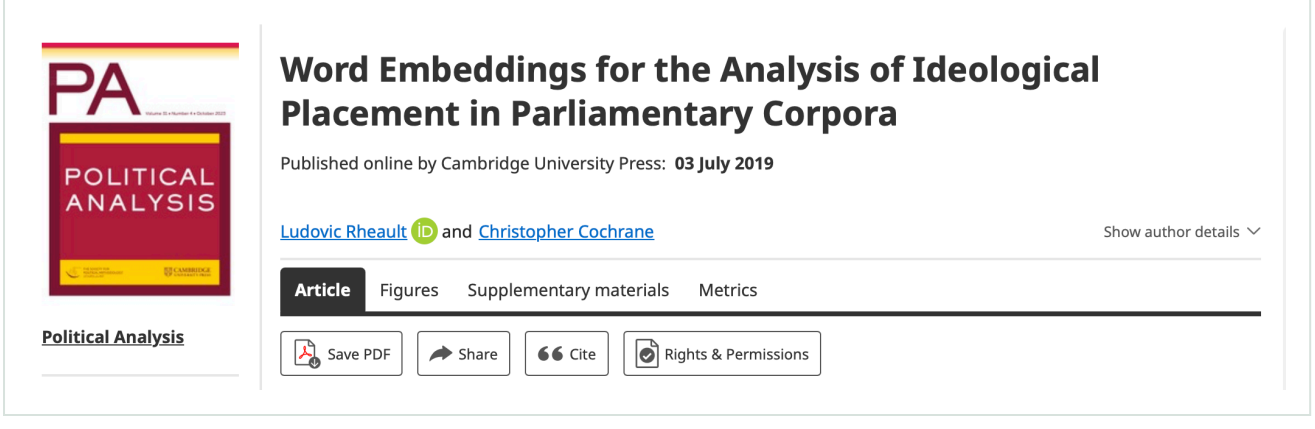
How four journal articles use embeddings

Case	Texts	How embeddings are used
Case 1 · Rheault & Cochrane	Parliamentary speech	Estimate party positions
Case 2 · Rodman	News about *equality*	Track a concept across eras
Case 3 · Müller & Proksch	European party manifestos	Find candidates for a nostalgia dictionary
Case 4 · Rodriguez et al.	U.S. Congressional Record	Compare party and gender differences in word use

We follow four studies, asking what gets a vector and how each result helps answer the research question.

Rheault & Cochrane (2020) · Rodman (2020) · Rodriguez et al. (2023) · Müller & Proksch (2024)

Case 1 · Party positions in parliament



Ludovic Rheault and Christopher Cochrane

Political Analysis, 2020 · Cambridge article page

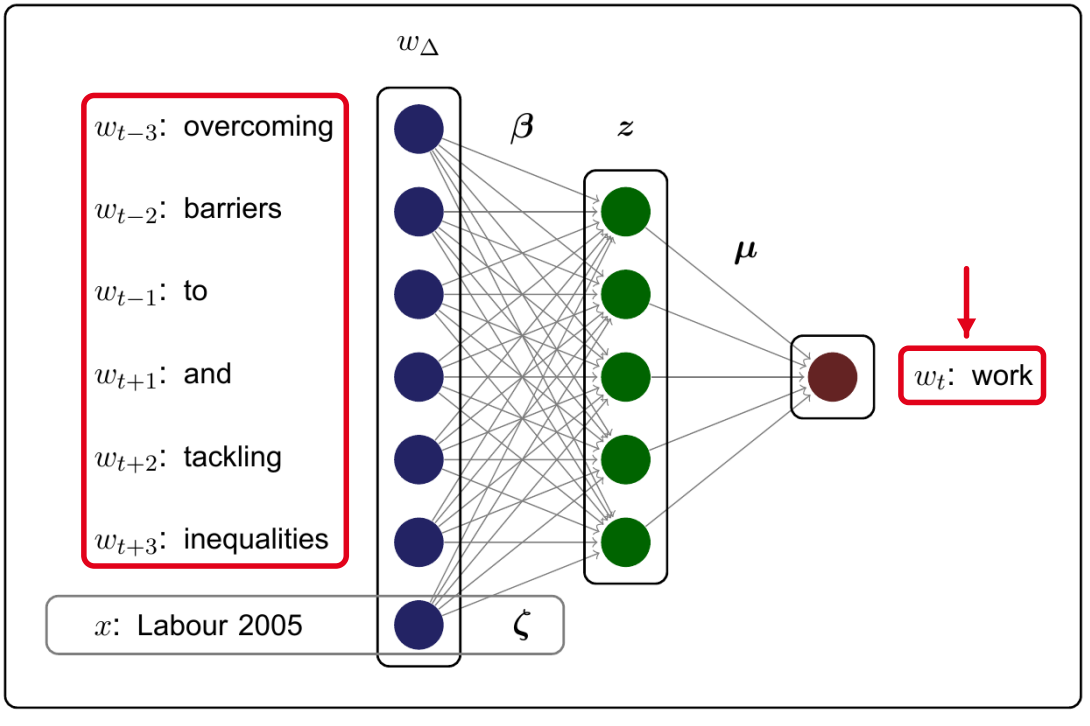
- **Background** Roll-call votes offer one way to locate parties on a political scale.
- **Question** Can speeches also reveal party positions and how they change between terms?
- **Texts** Parliamentary debates from the United States, Britain and Canada. The model represents each party in each legislative term.

Rheault & Cochrane (2020), *Political Analysis* 28(1), 112–133 · Author preprint

Learn party embeddings with political metadata

Case 1 · Party positions

How embeddings are used



1. Predict: The model predicts *work* from context and the party-term label.

2. Represent: Each party in each term receives a vector, here **Labour 2005**.

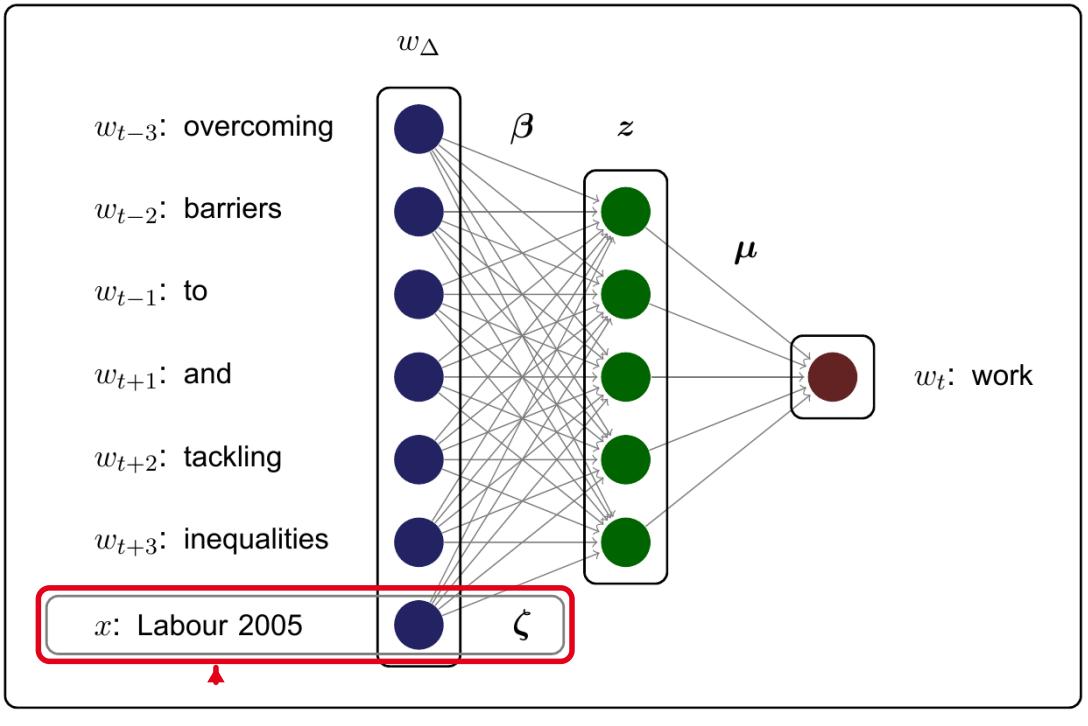
3. Project: We use PCA to compare the party-term vectors on PC1, shown next.

Rheault and Cochrane (2020), author preprint Figure 1, p. 8.

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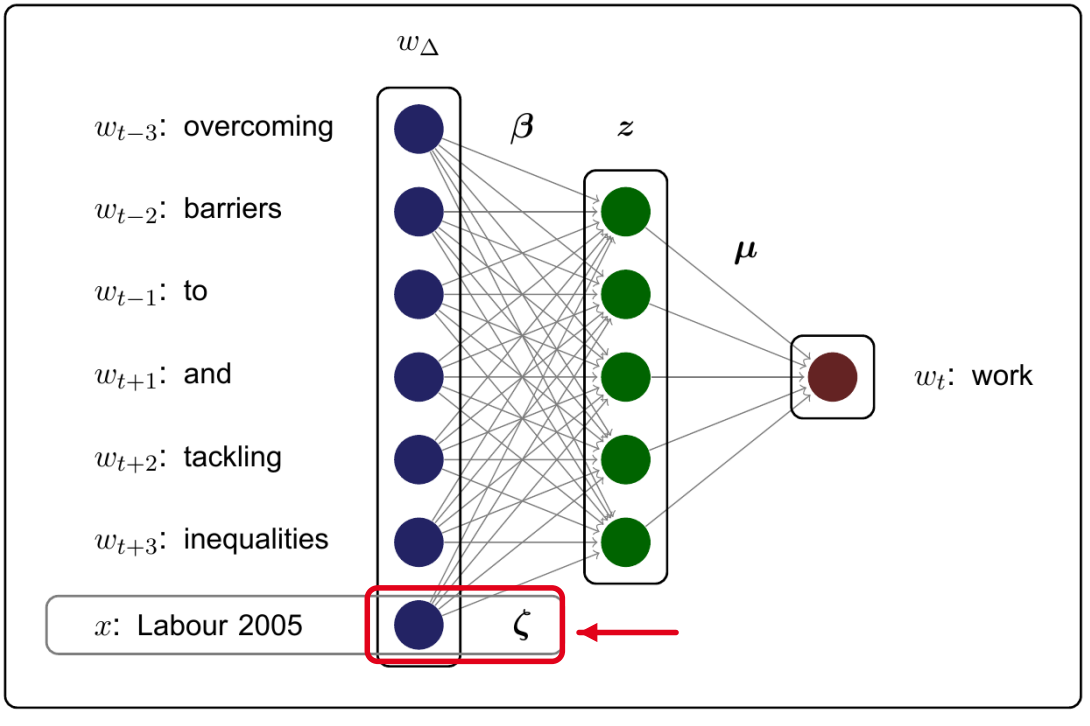
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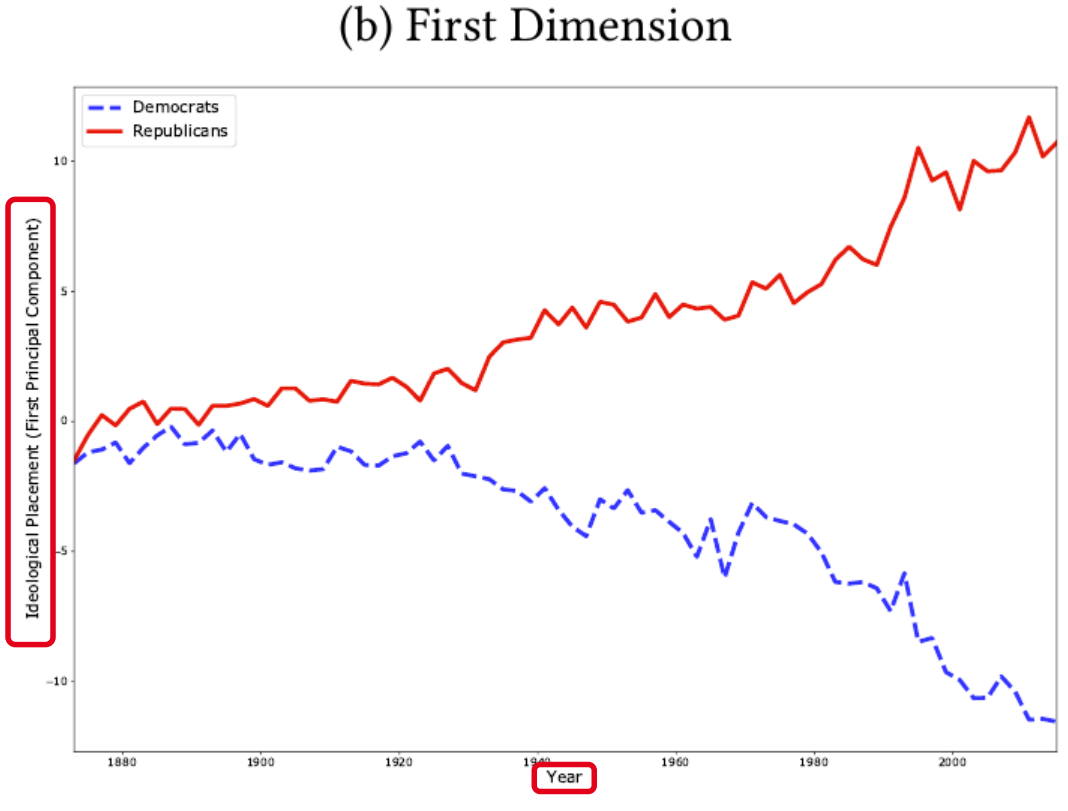
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From party vectors to a political dimension

Case 1 · Party positions



Read the axes: We compare years horizontally and PC1 vertically. PC1 captures the greatest variation across party-term vectors.

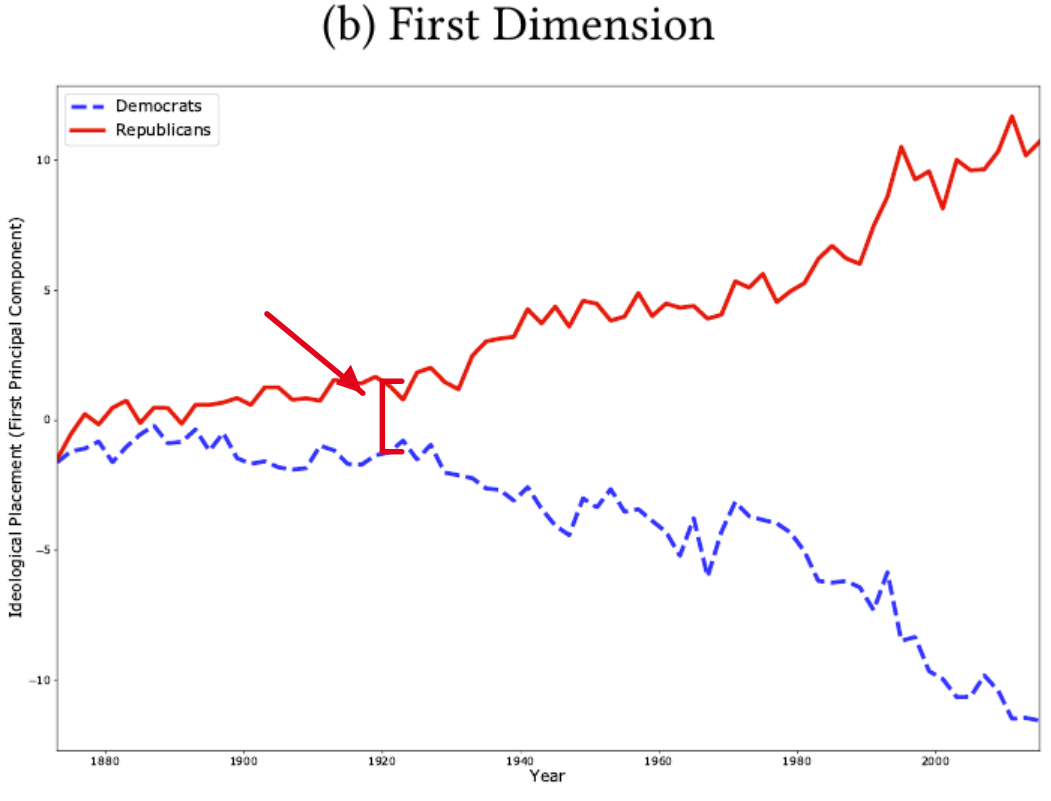
1920 bracket: We see a small gap: Democrats are dashed blue; Republicans solid red.

Answer: By **2000**, we see the parties much farther apart on this speech-based scale.

Rheault and Cochrane (2020), author preprint Figure 2b, p. 15.

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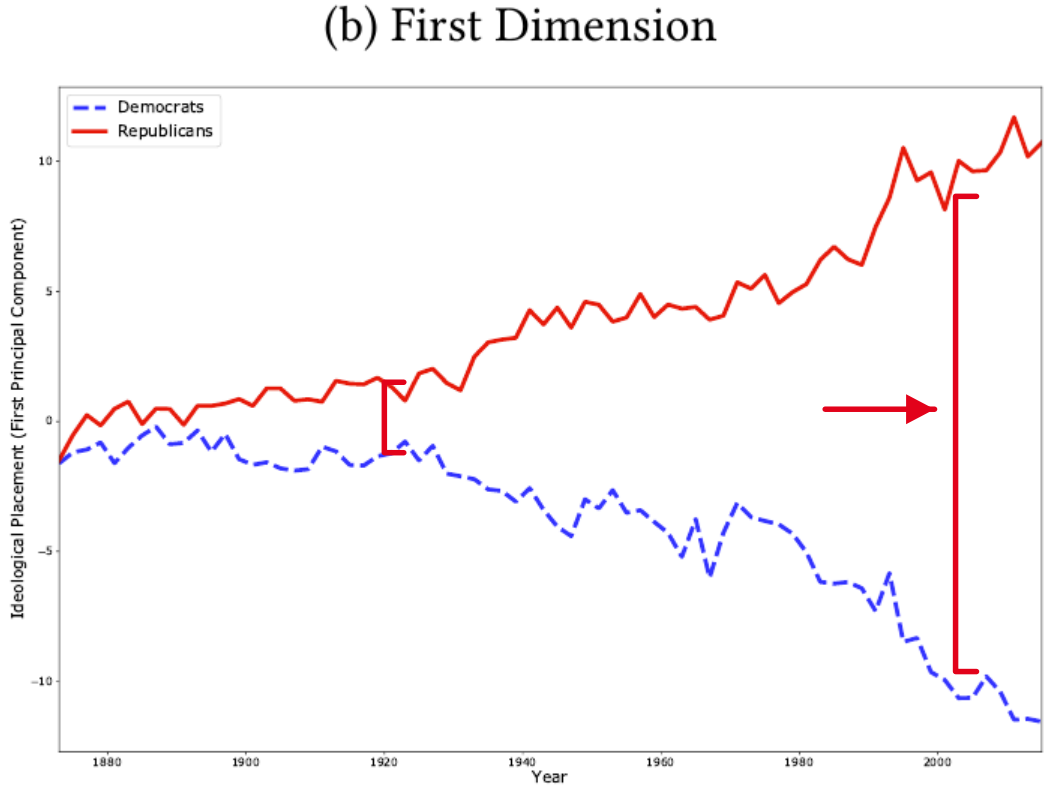
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Validate the positions outside the model

Case 1 · Party positions

Gold Standard	Metric	US House	US Senate	Canada	Britain
Voteview (1921-2016)	Correlation	0.918	0.919		
	Pairwise Accuracy	85.66%	83.93%		
Experts Surveys (1984-2002)	Correlation	0.982	0.981	0.869	0.910
	Pairwise Accuracy	86.67%	100.00%	87.88%	83.33%
rile (1945-2015)	Correlation	0.624	0.634	0.768	0.678
	Pairwise Accuracy	72.61%	72.25%	77.03%	74.98%
vanilla (1945-2015)	Correlation	0.736	0.736	0.731	0.755
	Pairwise Accuracy	75.62%	75.09%	76.30%	78.06%
legacy (1945-2015)	Correlation	0.898	0.907	0.855	0.876
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U.S. House Speech-based PC1 correlates **0.918** with Voteview party averages (1921–2016).

Next row: We read **85.66%** agreement on party-term pair orderings.

Answer: We can read PC1 as a left–right scale in this overlapping period. Agreement varies across benchmarks.

Rheault and Cochrane (2020), author preprint Table 2, p. 20.

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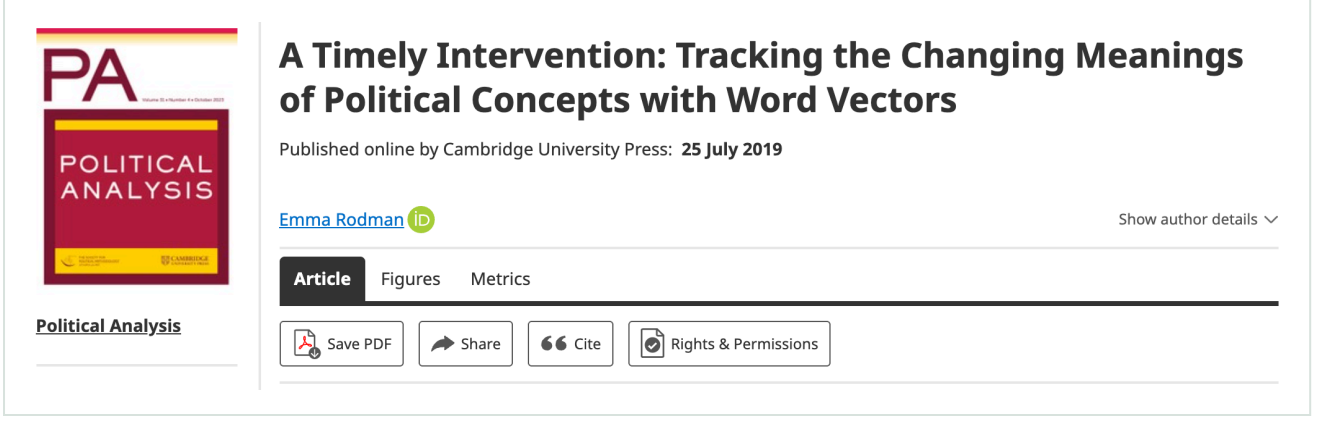
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Case 2 · The changing meaning of equality

Case 2 · Semantic change



Emma Rodman

Political Analysis, 2020 · Cambridge article page

- **Background** The word *equality* persists while debates about it change.
- **Question** Which groups and issues are associated with *equality* in different eras?
- **Texts** 3,105 news items from the New York Times Article API with *equality* in the headline, 1855–2016.

Rodman (2020), *Political Analysis* 28(1), 87–111 · Author manuscript

Newspaper contexts become historical evidence

Case 2 · Semantic change

How embeddings are used

Research step	What Rodman does
Collect contexts	Use headlines, abstracts and first paragraphs when available. Group the items into seven eras.
Estimate embeddings	Compare four Word2Vec training schemes. Each produces word vectors by era.
Track associations	Compute cosine between *equality* and selected words within each era. Follow those scores across eras.

Rodman (2020), author manuscript §§5.2, 5.4.

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Compare time-sensitive word vectors

Case 2 · Semantic change

	Deviance	Squared Deviance	Correlation
Naive Time Model	26.612	25.589	0.554
Overlapping Model	25.163	25.601	0.563
Chronologically Trained Model	**22.689**	**21.860**	**0.611**
Aligned Model	25.385	25.775	0.514

Read the columns: We look for lower deviations and higher correlation with the benchmark.

Compare the row: The chronological row has the highest correlation (**0.611**) and lowest deviations.

Choose a model: This model most closely matches five human-coded topic trends. Rodman then traces a new word pair.

Rodman (2020), author manuscript Table 2, p. 25.

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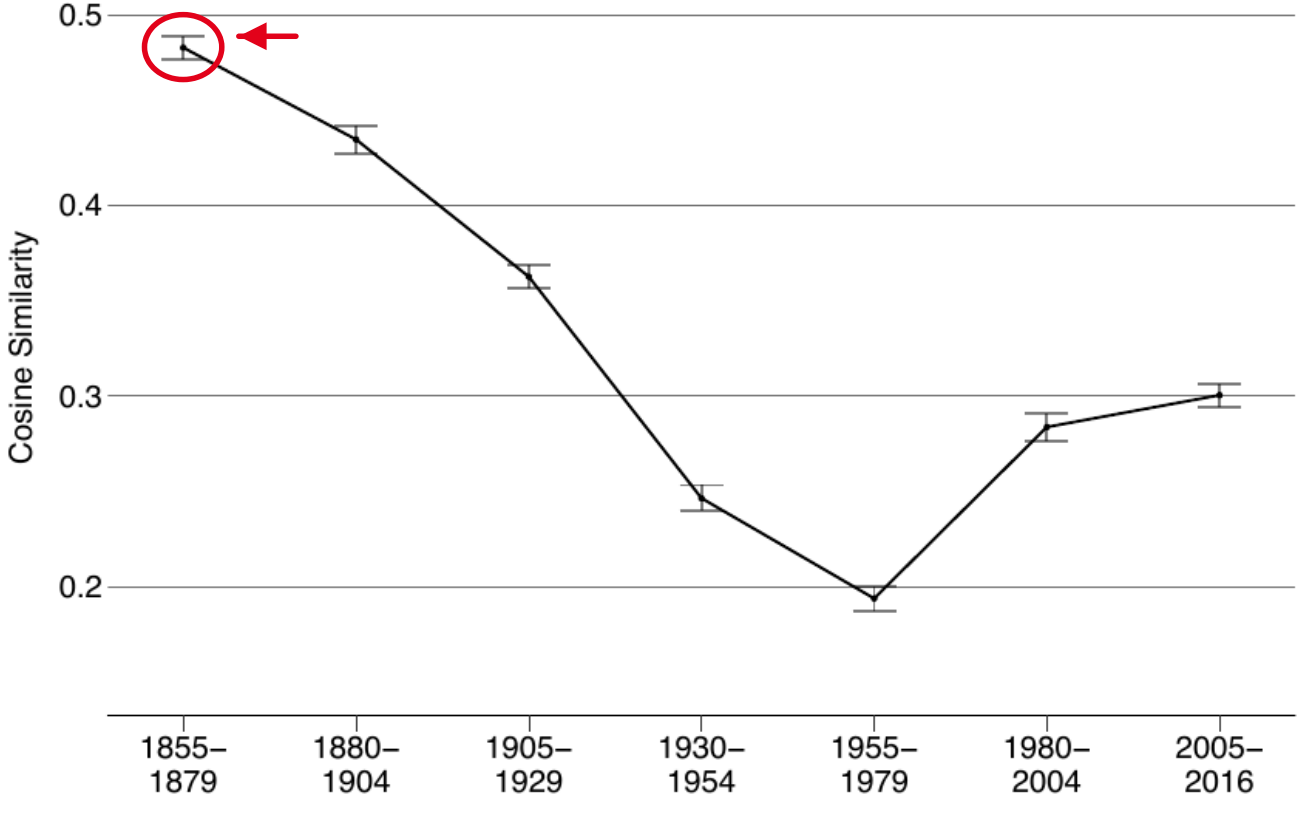
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Read semantic change with historical evidence

Case 2 · Semantic change



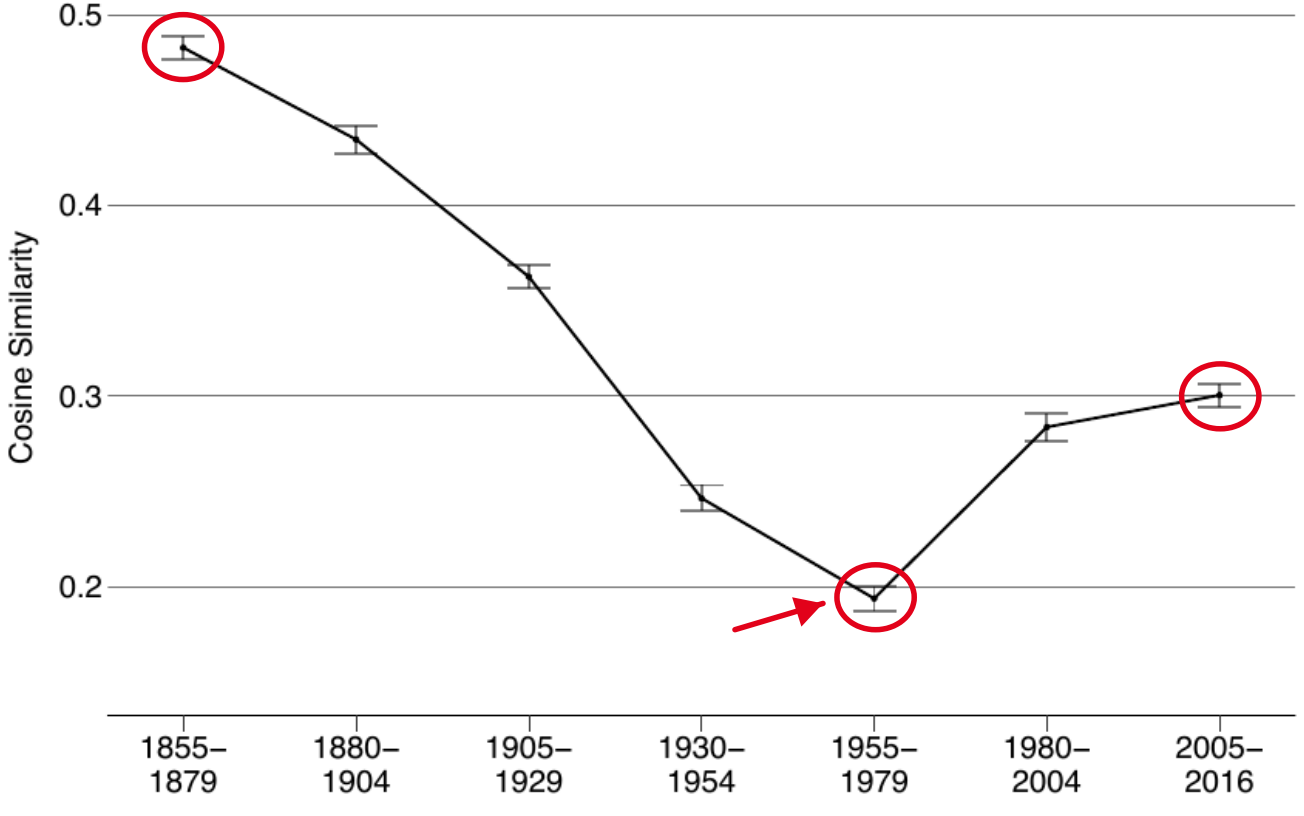
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Rodman (2020), author manuscript Figure 4, p. 29.

Read semantic change with historical evidence

Case 2 · Semantic change



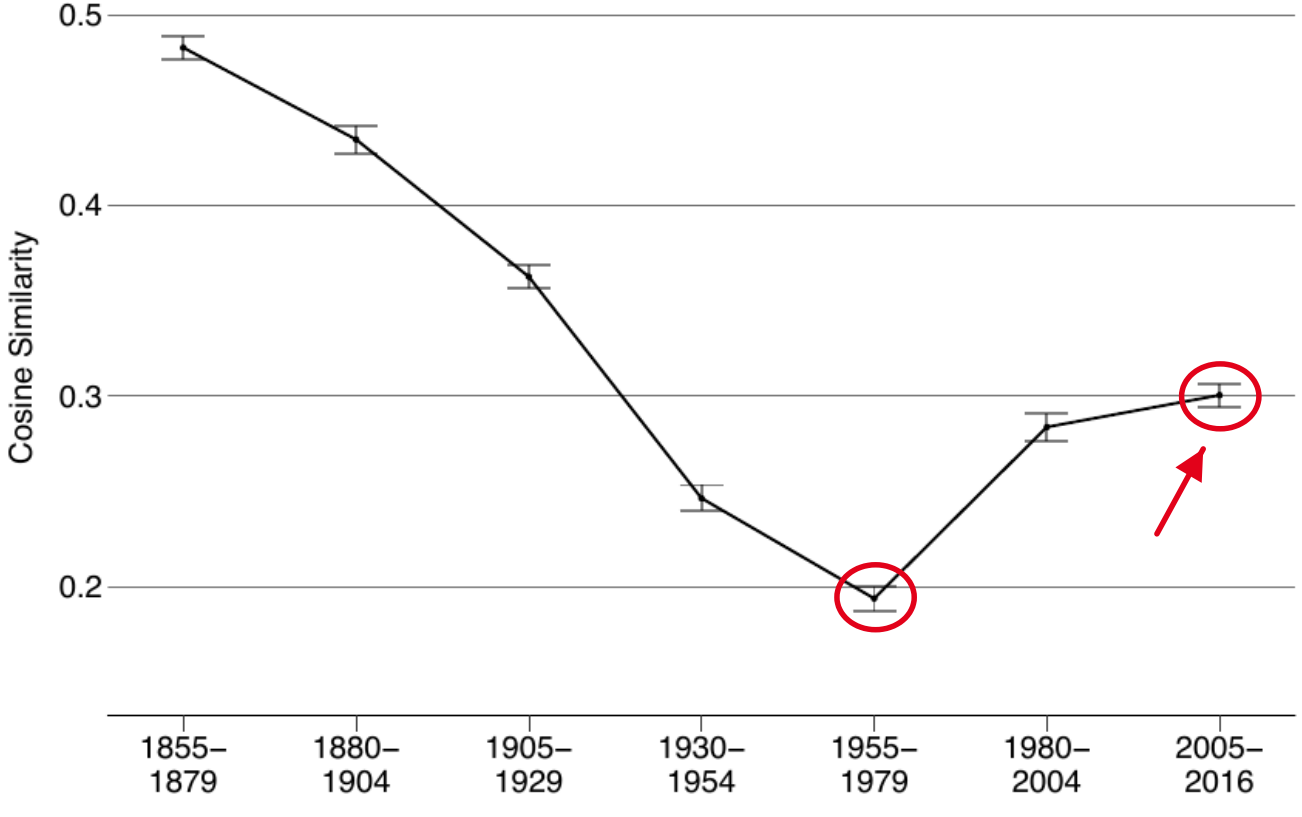
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Case 2 · Semantic change



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Discuss: We see a fall to **.19**, then a rebound to **.30**. Does the rebound show the earlier use continues?

Check passages Early uses could oppose interracial intimacy. Later news also connects *social equality* to social justice and institutional access.

Rodman (2020), author manuscript Figure 4, p. 29.

Case 3 · Nostalgia in party manifestos



Stefan Müller and Sven-Oliver Proksch

British Journal of Political Science, 2024 · Cambridge article page

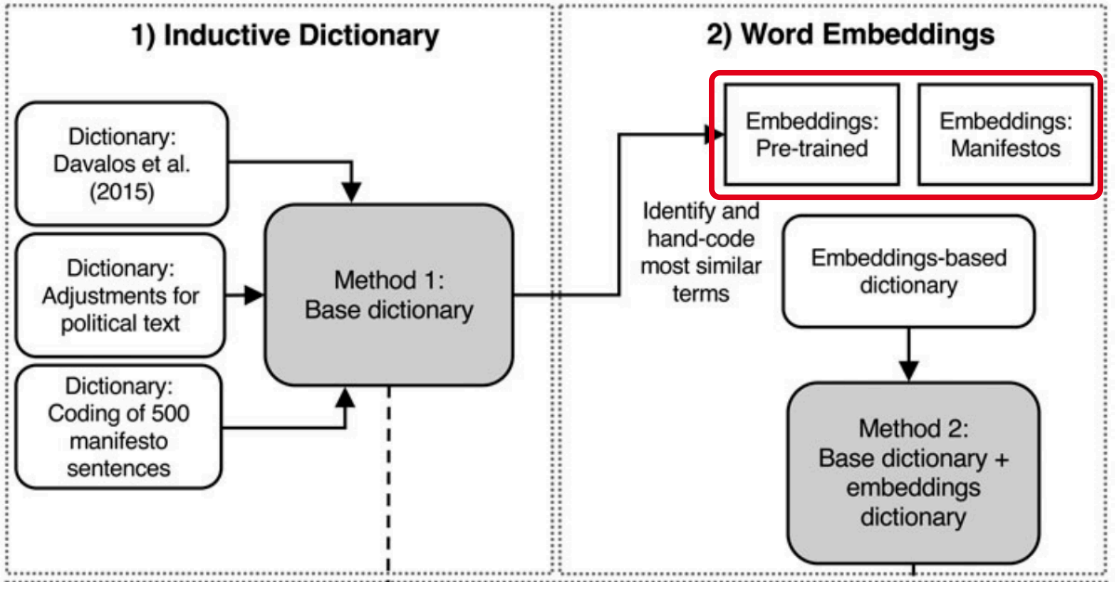
- **Background** A reference to the past can express praise, criticism or nostalgia.
- **Question** Do culturally conservative parties appeal more to a better past?
- **Texts** 1,648 manifestos from 24 European countries, 1946–2018.

W10 supplementary reading · Müller & Proksch (2024), *BJPS* 54(3), 993–1005

From word neighbours to a nostalgia measure

Case 3 · Nostalgia

How embeddings are used



1. Find candidates Search for neighbours of seed terms in pretrained GloVe and vectors trained on manifestos.

2. Screen candidates Three coders review the neighbours. Accepted terms form an embeddings-based dictionary.

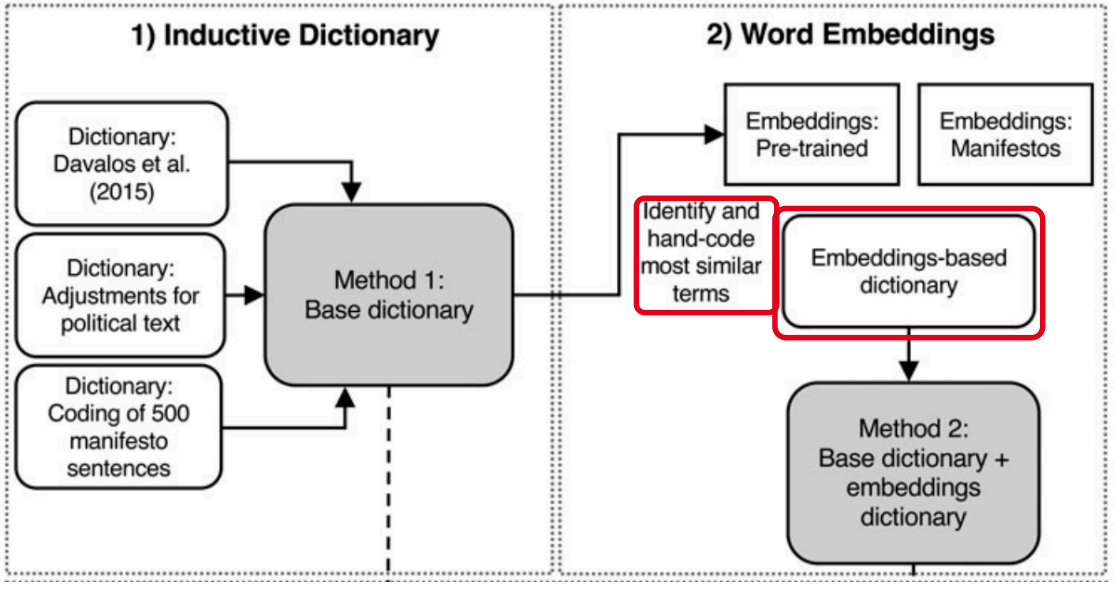
3. Score manifestos The bottom box combines both dictionaries. Flagged sentences per 1,000 become the outcome for comparing parties.

Müller & Proksch (2024), Figure 1, panels 1–2, p. 996 · Supplement, §B.2

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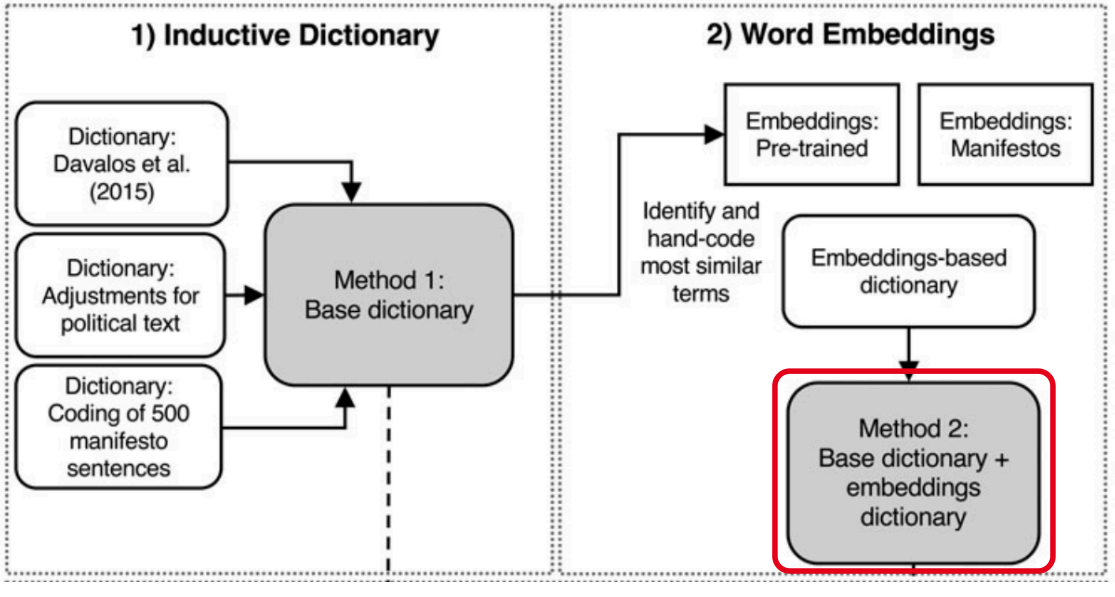
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Case 3 · Nostalgia

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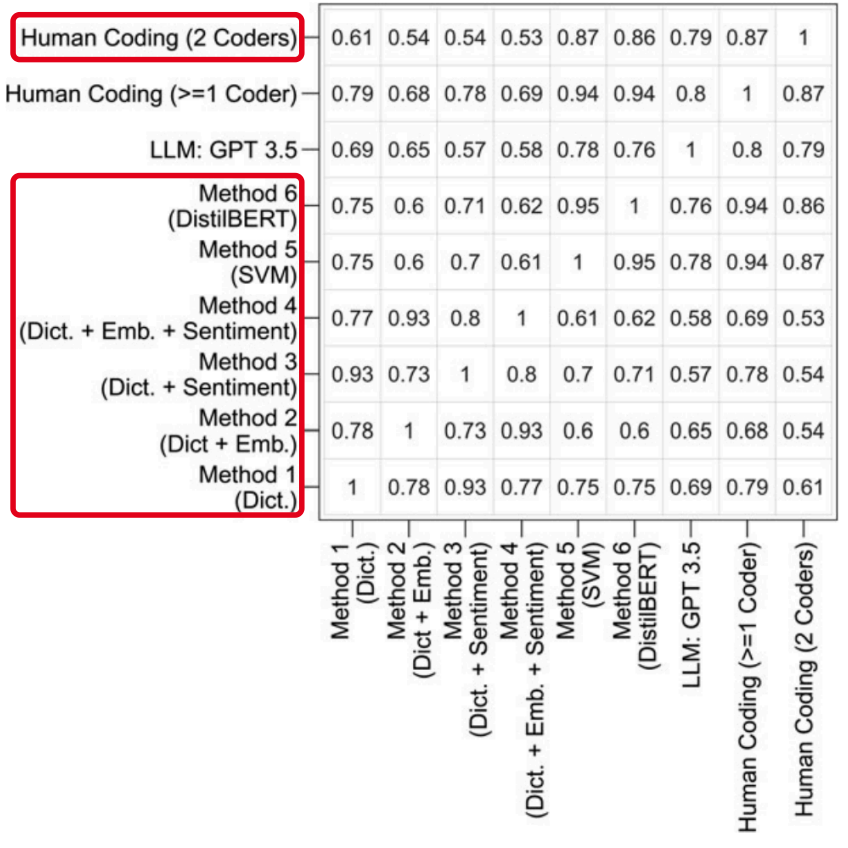
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Validate the measurement before comparing parties

Case 3 · Nostalgia



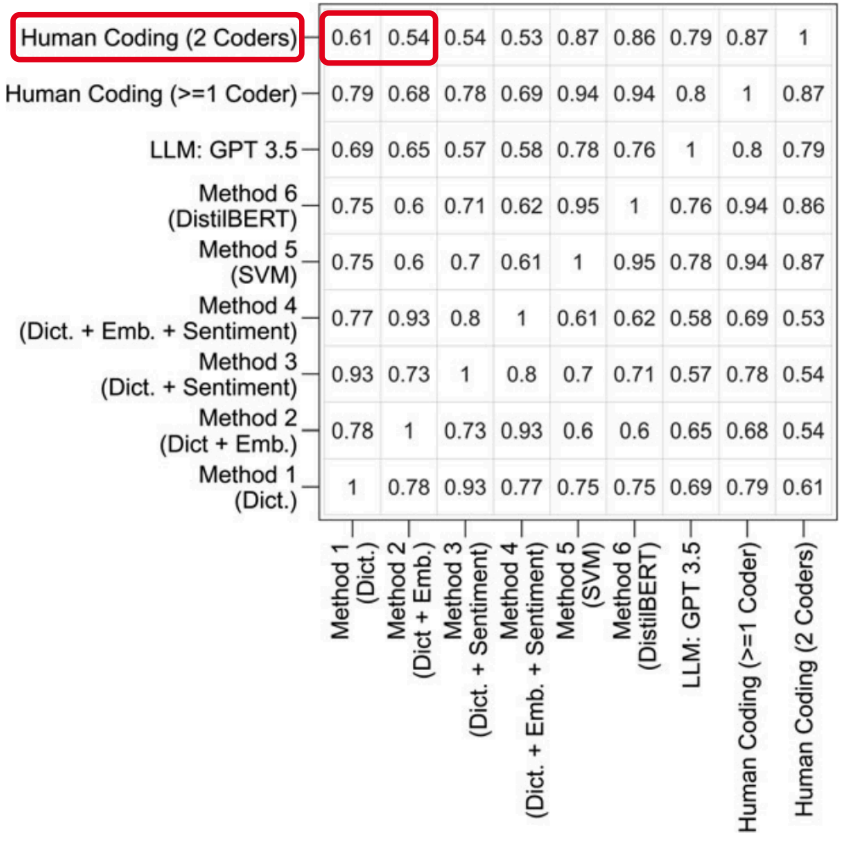
Methods. We compare manifesto scores from dictionaries (M1–M4) and sentence classifiers (M5–M6), using **50 manifestos**.

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Validate the measurement before comparing parties

Case 3 · Nostalgia



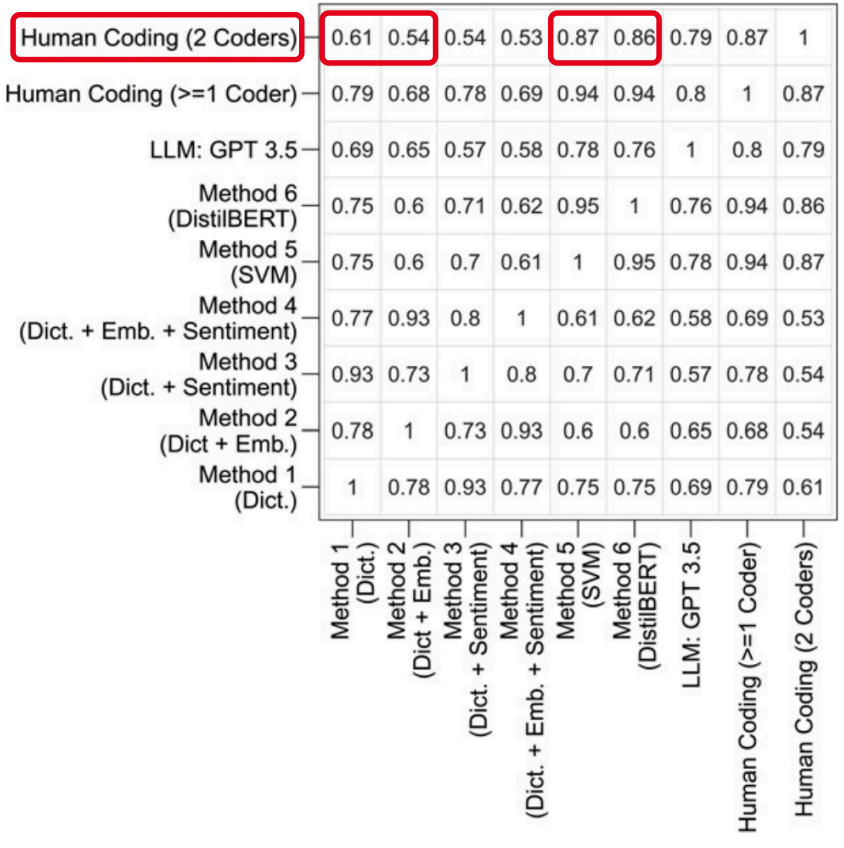
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Case 3 · Nostalgia



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Check. We see lower agreement after expansion. SVM: **.87**; DistilBERT: **.86**.

Müller & Proksch (2024), Figure 2, p. 998. Sentence validation on p. 997.

Compare parties across six measures

Case 3 · Nostalgia

Table 1. Predicting nostalgia for various measurements with standardized dependent variables

	Dictionary-Based Methods (M1–M4)				Machine Learning (M5–M6)	
	M1	M2	M3	M4	M5	M6
(Intercept)	0.02	−0.11	−0.01	−0.15	0.17	0.21
	(0.10)	(0.11)	(0.10)	(0.10)	(0.13)	(0.12)
Cultural Conservatism	0.10***	0.08***	0.08***	0.07**	0.07**	0.08**
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Government	0.12*	0.08	0.11*	0.13*	0.08	0.05
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Vote Share	0.00	0.00	0.00	0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Unemployment (*t*–1)	0.00	0.02**	0.01	0.02**	0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
AIC	4,157.19	4,062.81	4,273.12	4,187.26	4,135.66	4,081.72
BIC	4,205.52	4,111.14	4,321.45	4,235.59	4,184.00	4,130.05
Log Likelihood	−2,069.59	−2,022.40	−2,127.56	−2,084.63	−2,058.83	−2,031.86
N	1,588	1,588	1,588	1,588	1,588	1,588
N Groups: Parties	365	365	365	365	365	365
N Groups: Elections	260	260	260	260	260	260
N Groups: Countries	24	24	24	24	24	24

Linear mixed-effects models with random intercepts for countries, parties, and elections. Standard errors in parentheses. M1: Base dictionary; M2: Base dictionary + embeddings dictionary; M3: Base dictionary + positive sentiment; M4: Base dictionary + embeddings dictionary + positive sentiment; M5: Bag-of-words classifier (SVM); M6: Transformer-based classifier (DistilBERT). ***p < 0.001; **p < 0.01; *p < 0.05.

Columns use six standardized nostalgia scores. **M2** uses the expanded dictionary. **M4** also requires positive sentiment.

Boxed row. We read **.08** in M2 and **.07** in M4.

One point higher on cultural conservatism is associated with **.07–.10 SD** more manifesto nostalgia across the six models, adjusting for other variables.

Müller & Proksch (2024), Table 1, p. 1001 · Cultural conservatism scale, Supplement §C.6

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Same word, different contexts

Case 4 · Method intuition · 1/3

	The same word	Contextual clues and interpretation
→	**chips · UK use**	Ketchup, knife and fork suggest **hot fries.**
	chips · US use	A bag suggests **potato crisps.**

Meaning here: we compare patterns of use. These are constructed examples.

Embedding Regression: Non-Technical Explainer, §1

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Embedding Regression: Non-Technical Explainer, §1

A context becomes an ALC vector

Case 4 · Method intuition · 2/3

Step	What the model uses
→ **1 · Look up context words**	We use their vectors from one pretrained space.
2 · Average	$\mathbf{u}_i = \frac{1}{\lvert C_i \rvert} \sum_{w \in C_i} \mathbf{v}_w$ for this occurrence's surrounding words.
3 · Calibrate	$\mathbf{y}_i = A\mathbf{u}_i$. A learned linear transformation corrects the context average.

One use → one vector. Different uses can receive different representations in the same space.

Non-Technical Explainer, §2 · Rodriguez, Spirling & Stewart (2023), ALC method

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Case 4 · Method intuition · 2/3

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Non-Technical Explainer, §2 · Rodriguez, Spirling & Stewart (2023), ALC method

The context vector is the regression outcome

Observation: one use of chips. **Outcome:** its D-coordinate context vector.

$$\mathbf{y}_i = \boldsymbol{\beta}_0 + UK_i\boldsymbol{\beta}_{UK} + \boldsymbol{\epsilon}_i.$$

UK_i: US = 0, UK = 1. Both coefficients are D-coordinate vectors.

Read the result: we use coefficient length for the size of the difference, then neighbours and passages to interpret it. Permutations assess uncertainty.

Next: we replace English variety with party labels and study immigration.

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Non-Technical Explainer, §§3–5

Case 4 · How parties use immigration

Case 4 · Embedding regression



Pedro L. Rodriguez, Arthur Spirling and Brandon M. Stewart

American Political Science Review, 2023 · Cambridge article page

- **Background** Parties can mention *immigration* equally often while using it in different contexts.
- **Question** How does the use of the same word vary by party and gender?
- **Texts** U.S. Congressional Record, 111th–114th Congresses.

W9 core reading · Rodriguez, Spirling & Stewart (2023), *APSR* 117(4), 1255–1274

conText: from contexts to regression

Case 4 · Embedding regression

conText · Official Quick Start



Embedding regression

The above framework allows us to explore semantic differences over one grouping variable at a time. However, we often want to look at covariate effects while controlling for other covariates or indeed go beyond discrete covariates. In Rodriguez, Spirling and Stewart (2021) we show how a `la carte` embeddings can be used within a regression-framework that allows us to do just that. The corresponding package function is `conText::conText()`.

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```
# two factor covariates
set.seed(2021L)
model1 <- conText(formula = immigration ~ party + gender,
                  data = toks_nostop_feats,
                  pre_trained = cr_glove_subset,
                  transform = TRUE, transform_matrix = cr_transform,
                  jackknife = FALSE, confidence_level = 0.95,
                  cluster_variable = NULL,
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                  window = 6, case_insensitive = TRUE,
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```

```
fit <- conText::conText(
  immigration ~ party + gender,
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`toks`: text + metadata. `E` and `A`: a matched pair.

1 · One occurrence, one row. We keep context, party and gender. n uses give an $n \times D$ outcome matrix Y.

2 · Build the outcome vector. We average context-word vectors, excluding the target, then apply the common ALC transformation.

$$\mathbf{u}_i = \text{mean}_{w \in C_i} \mathbf{v}_w, \qquad \mathbf{y}_i = A\mathbf{u}_i.$$

3 · Regress all coordinates. R: Republican; M: male.

$$\mathbf{y}_i = \boldsymbol{\alpha} + R_i\boldsymbol{\beta}_R + M_i\boldsymbol{\beta}_M + \boldsymbol{\epsilon}_i.$$

$\mathbf{y}_i$, $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_R$ and $\boldsymbol{\beta}_M$ each have D entries.

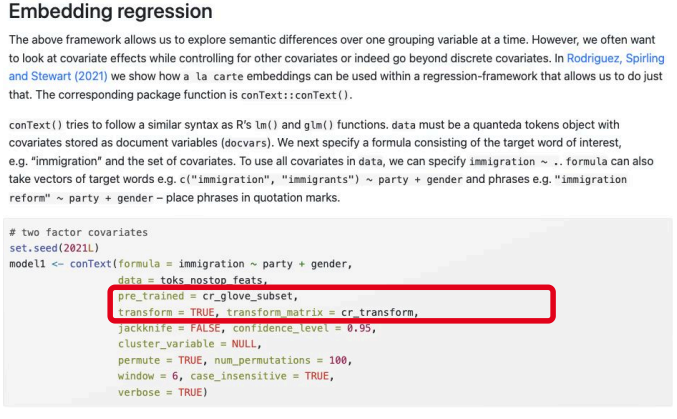
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Rodriguez, Spirling & Stewart (2023), pp. 1258–1259, 1264–1266 · Official conText Quick Start · After-class R walkthrough

conText: from contexts to regression

Case 4 · Embedding regression

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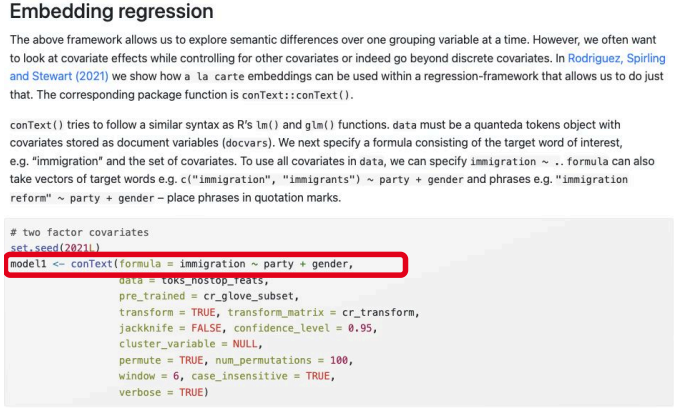
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# two factor covariates
set.seed(2021L)
model1 <- conText(formula = immigration ~ party + gender,
                  data = toks_nostop_feats,
                  pre_trained = cr_glove_subset,
                  transform = TRUE, transform_matrix = cr_transform,
                  jackknife = FALSE, confidence_level = 0.95,
                  cluster_variable = NULL,
                  permute = TRUE, num_permutations = 100,
                  window = 6, case_insensitive = TRUE,
                  verbose = TRUE)
```

```
fit <- conText::conText(
  immigration ~ party + gender,
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  transform = TRUE,
  transform_matrix = A)
```

`toks`: text + metadata. `E` and `A`: a matched pair.

1 · One occurrence, one row. We keep context, party and gender. n uses give an $n \times D$ outcome matrix Y.

2 · Build the outcome vector. We average context-word vectors, excluding the target, then apply the common ALC transformation.

$$\mathbf{u}_i = \text{mean}_{w \in C_i} \mathbf{v}_w, \qquad \mathbf{y}_i = A\mathbf{u}_i.$$

3 · Regress all coordinates. R: Republican; M: male.

$$\mathbf{y}_i = \boldsymbol{\alpha} + R_i\boldsymbol{\beta}_R + M_i\boldsymbol{\beta}_M + \boldsymbol{\epsilon}_i.$$

$\mathbf{y}_i$, $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_R$ and $\boldsymbol{\beta}_M$ each have D entries.

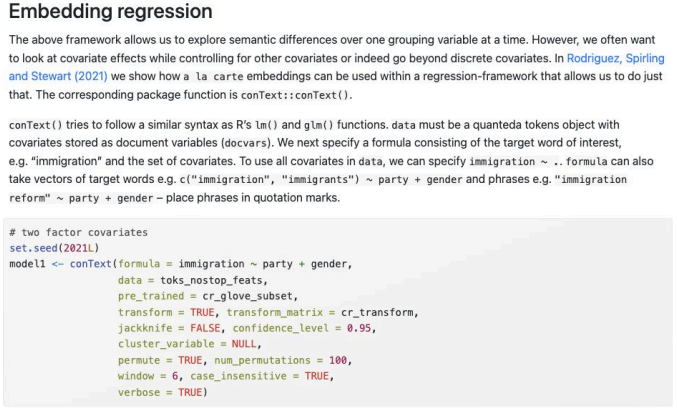
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Rodriguez, Spirling & Stewart (2023), pp. 1258–1259, 1264–1266 · Official conText Quick Start · After-class R walkthrough

conText: from contexts to regression

Case 4 · Embedding regression

conText · Official Quick Start



Embedding regression

The above framework allows us to explore semantic differences over one grouping variable at a time. However, we often want to look at covariate effects while controlling for other covariates or indeed go beyond discrete covariates. In Rodriguez, Spirling and Stewart (2021) we show how `a la carte` embeddings can be used within a regression-framework that allows us to do just that. The corresponding package function is `conText::conText()`.

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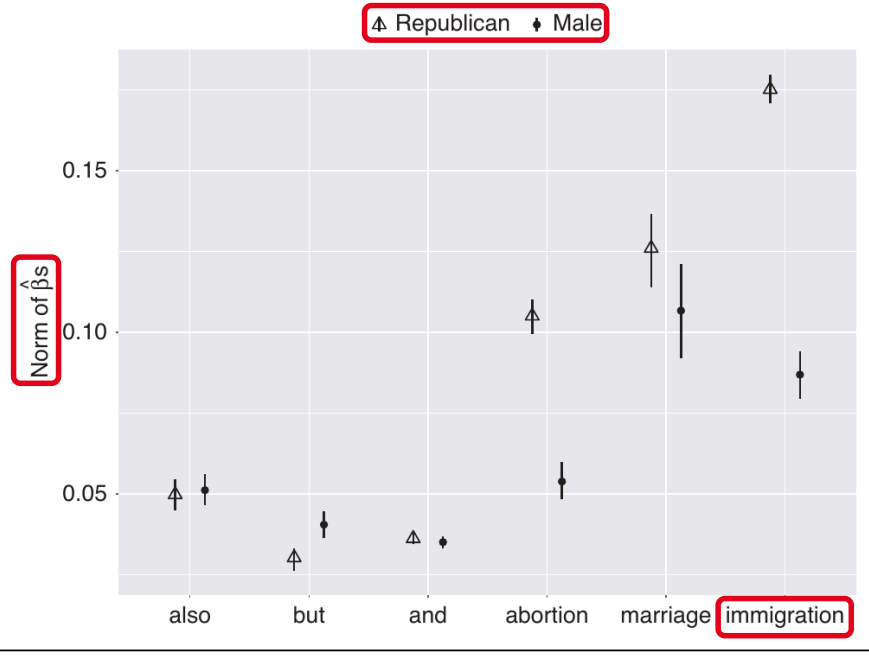
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Compare party and gender differences

Case 4 · Embedding regression

FIGURE 5. Differences in Word Meaning by Gender and Party



Note: Generally, different genders in the same party have more similar understanding of a term than the same gender across parties. See SM Section J for full regression output. All coefficients are statistically significant at 0.01 level.

Height. The adjusted difference vector's length: **0 = no vector difference**. Larger = greater difference. Bars: 95% bootstrap intervals.

Compare. For *immigration*, we read party **.175** at fixed gender and gender **.087** at fixed party. Both are vector lengths.

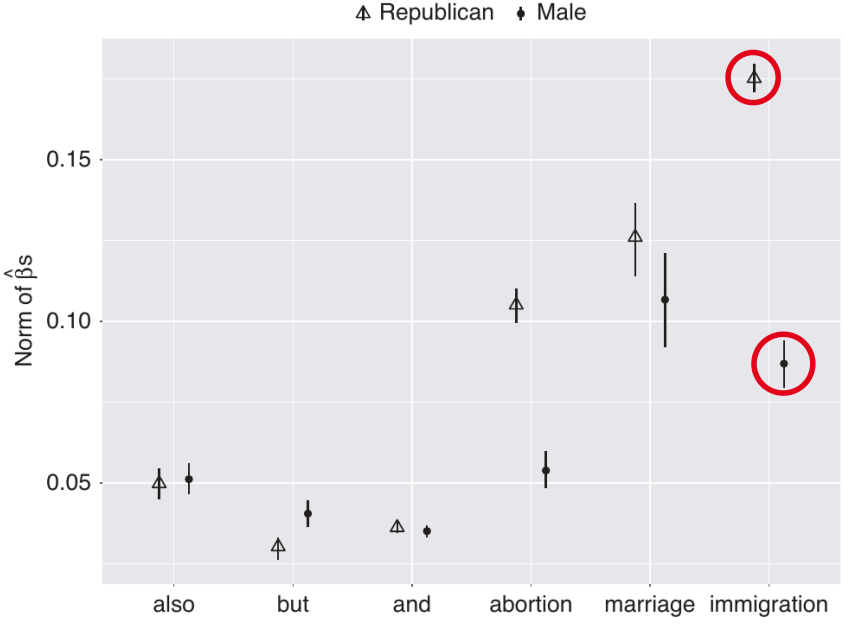
Answer. We see larger estimated usage gaps by party for all three political terms.

Rodriguez, Spirling & Stewart (2023), Figure 5 · Supplement Table J.4

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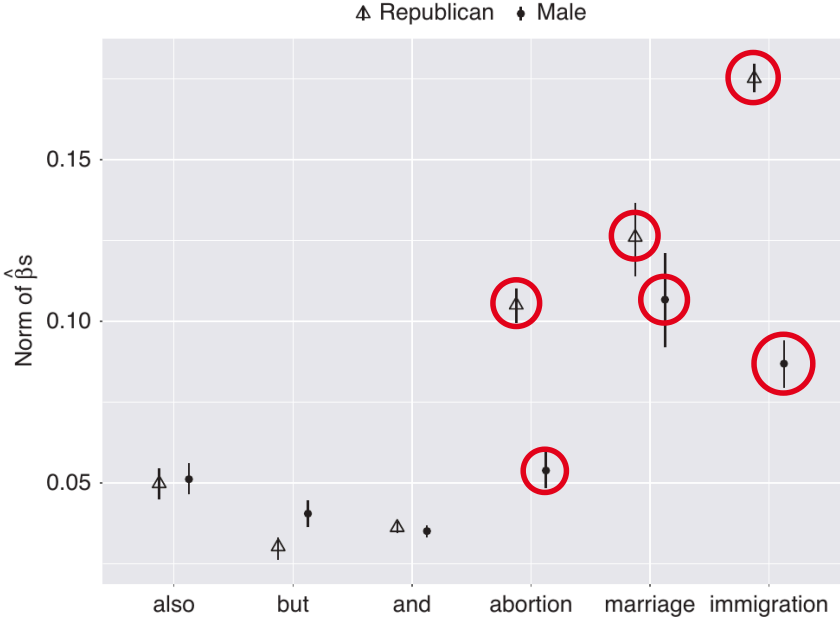
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What do group differences contain?

Case 4 · Embedding regression

TABLE 4. Top 10 Nearest Neighbors for the Target Term "Immigration"

Democrats	enact, overhauling, reform, legislation, enacting, overhaul, reforming, revamp, entitlement, bipartisan
Republicans	enforce, laws, enact, enacting, legislate, legislations, enforcing, regularize, immigration, legislation

Read each row. Each row lists the ten words closest by cosine to that party's estimated *immigration* vector.

Interpret the boxes. We see **changing the system** near Democrats and **enforcing existing law** near Republicans.

We also see *enact* and *legislation* in both rows. Next, we check the passages.

Rodriguez, Spirling & Stewart (2023), Table 4, p. 1267

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Return to the original contexts

TABLE 5. Subset of Top Nearest Contexts For The Target Term "Immigration"

Democrats

this congress to take on comprehensive `immigration` reform and fix our broken immigration

should get to work on comprehensive `immigration` reform the immigration system we have

Republicans

administration wants to ignore our nation's `immigration` laws and immigration process the problem

broken is the enforcement of our `immigration` laws and we have seen that

Read the party labels. We read passages close to each party's estimated vector.

Answer. We see **reform** in the Democratic passage and **enforcement** in the Republican passage.

Scope. These passages illustrate the contrast. To estimate frequency, we need a broader sample.

Rodriguez, Spirling & Stewart (2023), Table 5, p. 1267

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Case 4 · Embedding regression

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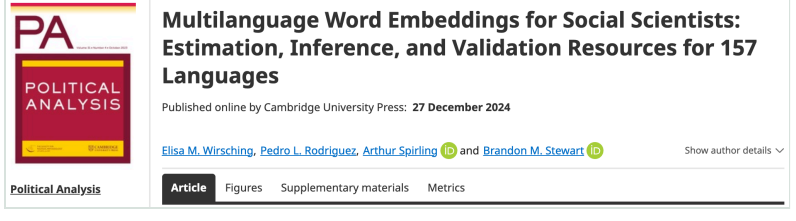
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Embedding regression beyond English

Case 4 · Multilingual extension



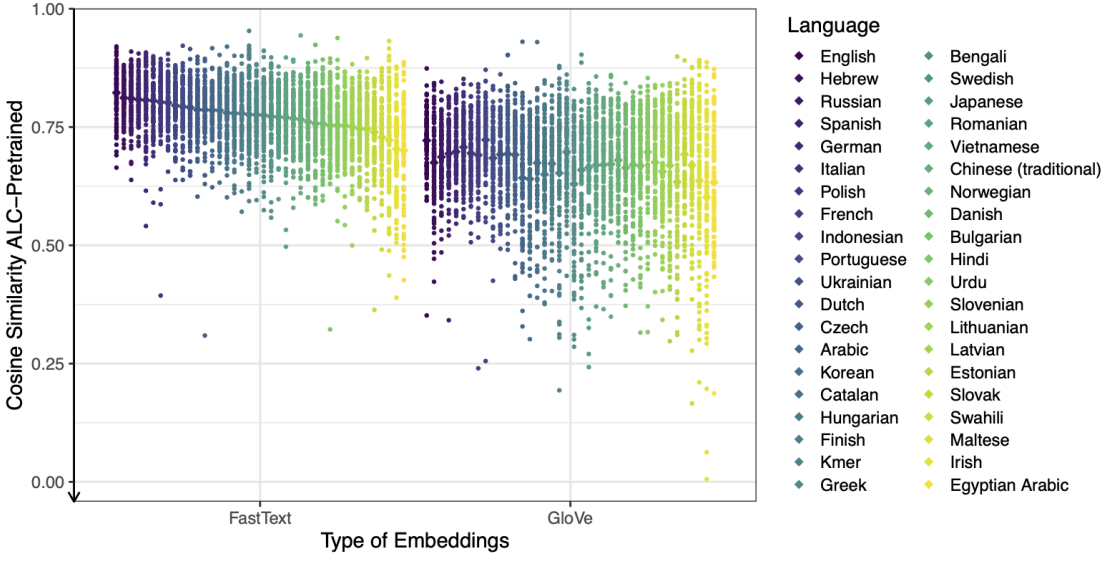


Figure 1. Reconstruction performance: cosine similarity between our ALC version of `fastText` and `GloVe` and those underlying architectures. Languages are ordered according to the mean accuracy for `fastText`. In theory, cosine similarities range between −1 and 1, but empirically all estimates are positive.

We compare ALC estimates with pretrained vectors within each language.

Wirsching et al. (2025), *Political Analysis*, Figure 1, p. 159

Four studies using embedding regression

Case 4 · Further applications


SPSR Swiss Political Science Review
Schweizerische Zeitschrift für Politikwissenschaft
Revue Suisse de Science Politique
Rivista Svizzera di Scienza Politica

RESEARCH NOTE | Open Access

How Populists Rally Around the People: Evidence from the Yellow Vests Movement Using an Embedding Regression Model

Frédéric Gonthier

First published: 27 January 2025 | https://doi.org/10.1111/spsr.12649 | VIEW METRICS

Gonthier (2025) · SPSR
French *peuple* across ideological groups.


PSJ POLICY STUDIES JOURNAL

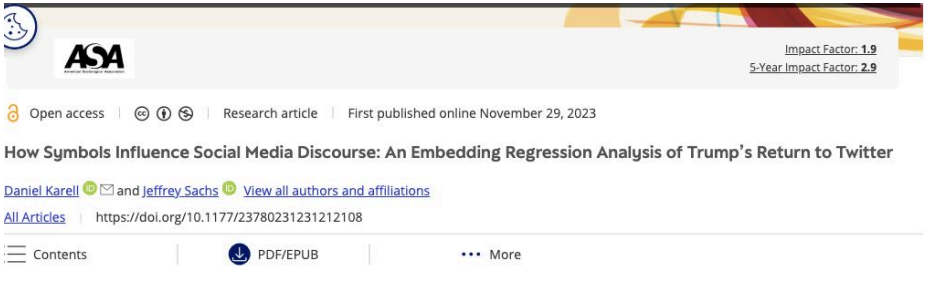
ORIGINAL ARTICLE | Open Access

A Natural Language Processing Approach to Identifying Partisan Framing of Climate Change Denialism, Fatalism, and Solutions in US Congressional Speeches

Joseph Charles Van Matre, Isabel L. Krakoff, Allegra H. Fullerton

First published: 08 October 2025 | https://doi.org/10.1111/psj.70079 | VIEW METRICS

Van Matre et al. (2026) · PSJ
Climate frames by party and congressional term.


ASA

Impact Factor: 1.9
5-Year Impact Factor: 2.9

Open access | Research article | First published online November 29, 2023

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Daniel Karell and Jeffrey Sachs View all authors and affiliations

All Articles | https://doi.org/10.1177/23780231231212108

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Minority-group language before/after a signal.


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Published 2021-10-28

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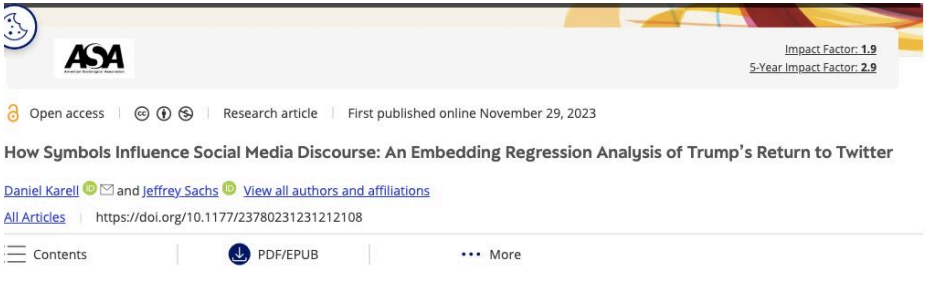
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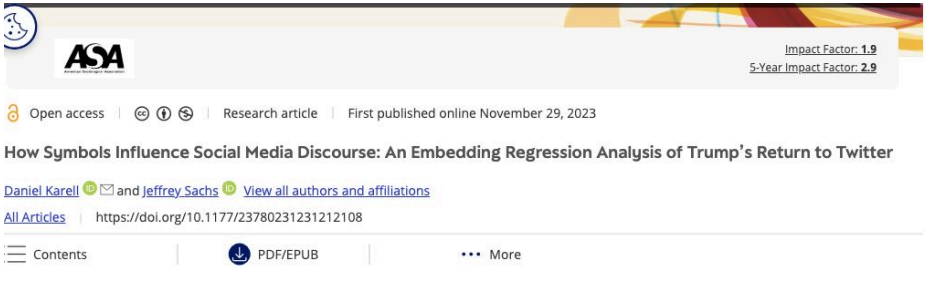
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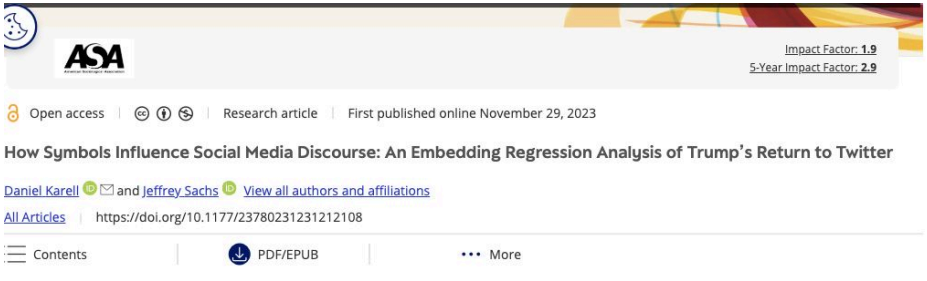
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Lab guide · Know the workflow before the break

Guide 1/3 · Goal and plan · 17:31–17:32

Goal: We learn what a vector represents, how to compare it and how to check our interpretation.

Classroom block	What the vector represents	Short Check
17:40–17:55 · Our 2024 Lab	A static vector for each word type; then a PCA projection.	**E06:** matrix units and identical security vectors. **PCA Check:** what the map leaves out.
17:55–18:10 · conText	A vector for each occurrence's context; then a conditional group comparison.	**E04:** unit → word/context evidence → party coefficient.

We read a short Code example, examine its Output and answer a Check. The Lab includes the full runnable steps.

Lab guide · Know the workflow before the break

Guide 2/3 · Our 2024 Flair/PCA workflow · 17:32–17:33:30

Step	Read the input and output	What it tells us
1 · Load	R/Python environment + cached GloVe → **100 coordinates**.	We load a trained model.
2 · Get vectors	Eleven listed words → **11 × 100** matrix.	One row per token in the word list.
3 · Compare security	Three sentences → **3 × 100** security matrix; cosine **1**.	Static lookup returns the same word-type vector.
4 · Read PCA	The same eleven vectors → **table + 2D + 3D**; PC1/2 retain **38.24%**.	We distinguish this projection from full-space cosine.

E06: explain the two matrix sizes and cosine 1. **PCA Check:** does proximity on the map guarantee full-space similarity?

Lab guide · Know the workflow before the break

Guide 3/3 · conText workflow · 17:33:30–17:35

Step	Read the input and output
C00 · Inputs	**400 texts** + **E: 500 × 300** word vectors + matched **A: 300 × 300**.
C01 · Contexts	Each immigration occurrence and its metadata → **924 context rows**.
C02 · ALC	Surrounding words + E/A → **924 × 300** context vectors → **2 × 300** party means.
C03–04 · Interpret	We check our reading of nearby words against retrieved context words.
C05 · Regression	We return to occurrences and compare **R with D, holding gender fixed**.

E04: How do we connect an observation, its word/context evidence and the party coefficient? Regression adds adjustment for gender.

Next: we take five minutes and begin Flair at **17:40**.

Five-minute break

We return at 17:40 for Flair preparation, with the Lab handout beside the slides.

30-minute Lab: Our 2024 Flair exercise and plots (15 min) → conText Quick Start (15 min).

We read each core call, examine its output and answer a short Check.

After class: full PCA computation and advanced extensions.

Lab · Prepare Flair

17:40–17:42. We check the R/Python connection and cached model.

Code

Before class, we prepare Python, packages and GloVe with Lab F00. From the course root:

```
source("W9/lab/flair/flair_prepare.R")
```

If R is unavailable, we use the Python route. Installation and downloads happen before class.

Output

Route	Passing checkpoint
R	Import check passed; cached model path
Python	Imports OK. Device: CPU.; cached model path

Ready to load the model. We have checked the connection and file path.

R preparation · Python preparation · Before-class F00

Lab · Flair 1: Load the model

17:42–17:44. We load GloVe in R or Python and check its dimensions.

Code

We stay in the same session. `glove_path` points to the cached model file.

```
embedding <- WordEmbeddings(glove_path)
print(embedding$embedding_length)
```

Output

→ **Model loaded · 100 coordinates.** `embedding` holds the pretrained GloVe model. Next, we attach its vectors to tokens.

Lab · Flair 2: Get word vectors

17:44–17:47. We create a Sentence, attach embeddings and read the coordinates. Python follows the same steps.

Code

```
word_list <- paste("King Queen man woman Paris
London",
  "apple orange Taiwan Dublin Bamberg")
word_sentence <- Sentence(word_list)
embedding$embed(word_sentence)
word_vectors <- do.call(rbind,
lapply(word_sentence$tokens,
  function(token)
as.numeric(token$embedding$tolist())))
print(dim(word_vectors))
```

Output

→ **Matrix · 11 × 100.** Each row is one token from the displayed word list; each column is a GloVe coordinate.

We use `Sentence()` to split the input, `embed()` to attach vectors and `tolist()` to read their numbers.

Lab · Flair 3: Compare security

17:47–17:52. We compare three occurrences, then explain the result in E06 at 17:50–17:52.

Code

We use the three sentences from the Lab. This call attaches vectors to their tokens.

```
embedding$embed(sentences)
```

Output

→ **Matrix · 3 × 100.** We keep one security vector from each sentence.

→

Constructed sentence	Static security vector
Social security supports households.	Same 100 coordinates
The United Nations Security Council met.	Same 100 coordinates
Energy security matters for jobs.	Same 100 coordinates

Each security-pair cosine is **1**.

→ **E06 · 17:50–17:52.** Explain both matrix sizes and why the security vectors are identical. If running the code, also note your route and versions.

Lab · PCA: Read the 2024 word-vector map

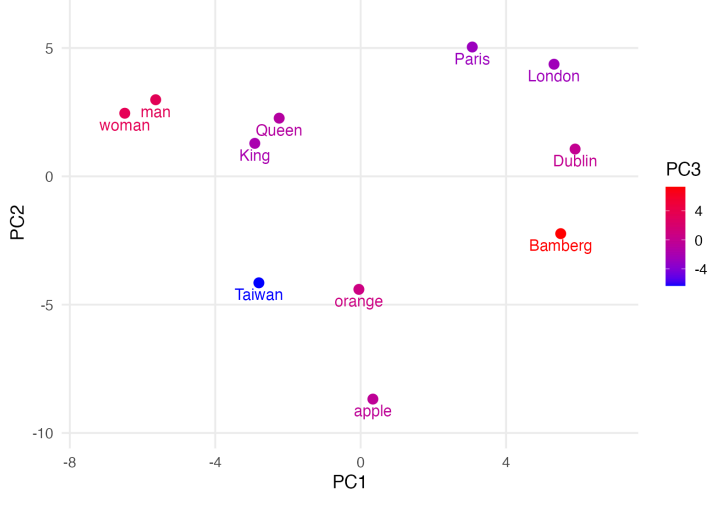
17:52–17:55. We read the 2D/3D figures now and leave the full computation for after class.

Code

Core call · full steps are in the Lab.

```
pca_result <-
prcomp(word_vectors,
  center = TRUE, scale. =
TRUE)
round(pca_result$x[1:3, 1:2],
2)
```

Output



One point = one word.

Word	PC1	PC2
King	−2.91	1.29
Queen	−2.24	2.27
man	−5.64	2.98

PC1 + PC2 retain **38.24%** of the standardized variance.

Lab · C00: Text, vectors and a matched transformation

17:55–17:58. We now use conText, starting with its texts, vectors and matching transformation.

Code

Core call · full steps are in the Lab.

```
print(c(documents = ndoc(cr_sample_corpus)))
print(rbind(E = dim(cr_glove_subset),
  A = dim(cr_transform)))
```

Output

Object	Recorded input
`cr_sample_corpus`	400 speeches with metadata
`cr_glove_subset`	500 words × 300 coordinates
`cr_transform`	Matching 300 × 300 transformation

Check: Which object contains the text? Which two must be a matched pair?

Lab · C01: Keep one occurrence per row

17:58–18:00. We extract immigration contexts and keep their party and gender labels.

Code

Core call · full steps are in the Lab.

```
immig_toks <- tokens_context(toks_nostop_feats,
  pattern = "immigration", window = 6L,
valuetype = "fixed",
  rm_keyword = TRUE)
```

Output

924 contexts · D: 486 · R: 438.

Pattern	Party	Gender
immigration	D	F
immigration	D	F
immigration	D	F

Check: One row is one use of immigration, carrying its speech metadata.

Core call and full step · Recorded output · Official Quick Start

Lab · C02: Turn contexts into ALC vectors

18:00–18:03. We turn each context into an ALC vector, then average by party.

Code

Core call · full steps are in the Lab.

```
immig_dfm <- dfm(immig_toks)
immig_dem <- dem(immig_dfm, pre_trained =
cr_glove_subset,
  transform = TRUE, transform_matrix =
cr_transform,
  verbose = FALSE)
immig_wv_party <- dem_group(immig_dem,
  groups = immig_dem@docvars$party)
```

Output

→ **DEM · 924 × 300.** One ALC context vector per row.

→ **Party means · 2 × 300.** We average contexts separately for D and R.

Check: We still have 300 coordinates. What does a row represent now?

Core call and full step · Recorded output · Official Quick Start

Lab · C03: Read the nearby words

18:03–18:05. We use nearby words to interpret the party averages.

Code

Core call · full steps are in the Lab.

```
immig_nns <- nns(immig_wv_party,
  pre_trained = cr_glove_subset, N = 3,
  candidates = immig_wv_party@features, as_list
= FALSE)
```

Output

→ **D · cosine neighbours**
immigration (0.848), broken (0.680), comprehensive (0.661)

→ **R · cosine neighbours**
immigration (0.875), illegal (0.708), enforcement (0.645)

Check: What difference in emphasis do we see? Next, we check a context.

Lab · C04: Read the nearby contexts

18:05–18:07. We check that interpretation against the nearby contexts.

Code

Core call · full steps are in the Lab.

```
immig_ncs <- ncs(immig_wv_party, contexts_dem =
immig_dem,
  contexts = immig_toks, N = 1, as_list =
FALSE)
```

Output

Near the D mean · cosine 0.802
→ end conversation passed immigration bill comprehensive bill clean broken immigration system year

Near the R mean · cosine 0.798
→ america suggest immigration can good thing illegal immigration legal immigration mean means

Check: We retrieved processed context words from a pool containing both parties.

Lab · C05: Introduce embedding regression

18:07–18:10. We return to occurrences and compare parties while holding gender fixed.

Code

Core call · full steps are in the Lab.

```
set.seed(2021L)
fit <- conText(immigration ~ party + gender,
  data = toks_nostop_feats, pre_trained =
cr_glove_subset,
  transform = TRUE, transform_matrix =
cr_transform,
  jackknife = FALSE, permute = TRUE,
  num_permutations = 100, window = 6L,
  valuetype = "fixed", verbose = FALSE)
```

Output

Coefficients · 3 × 300. Intercept, party_R and gender_M. Unlike the earlier party means, `party_R` adjusts for gender.

Coefficient	Debiased squared norm	Permutation p
party_R	8.668	0
gender_M	5.155	0

`0` records 0 exceedances in 100 permutations.

E04 · Check: What does party_R compare at fixed gender? How can we explain the result using a neighbour and a context?

Design an embedding study in one paragraph

1. We start with a political question and the **unit** that receives a vector.
2. We choose the corpus, representation and comparison for that question.
3. We identify evidence that could change our interpretation.

Our design. How might we compare a policy term across parties, periods or legislative settings?

Text as Data, §§8.2 and 8.4 · W9 case guide

Readings and classroom materials

Concepts and demonstrations

- Grimmer, Roberts & Stewart (2022), *Text as Data*, Ch. 8. Ch. 7 covers similarity and TF–IDF.
- Schoonvelde (2026), ESS Day 3, *Comparing Representations*.

Assigned studies

- Bischof & Senninger (2018), campaign language and political knowledge.
- Rodriguez & Spirling (2022), evaluating word embeddings.
- Rodriguez, Spirling & Stewart (2023), embedding regression.

References

W9 assigned readings

Bischof & Senninger (2018) · *EJPR* · DOI
Simple Politics for the People? Complexity in Campaign Messages and Political Knowledge

Rodriguez & Spirling (2022) · *JOP* · DOI
Word Embeddings: What Works, What Doesn't, and How to Tell the Difference for Applied Research

Rodriguez, Spirling & Stewart (2023) · *APSR* · Case 4 · DOI
Embedding Regression: Models for Context-Specific Description and Inference

W9 optional reading

Wirsching et al. (2025) · *Political Analysis* · DOI
Multilanguage Word Embeddings for Social Scientists: Estimation, Inference, and Validation Resources for 157 Languages

Additional W9 cases

Rheault & Cochrane (2020) · *Political Analysis* · Case 1 · DOI
Word Embeddings for the Analysis of Ideological Placement in Parliamentary Corpora

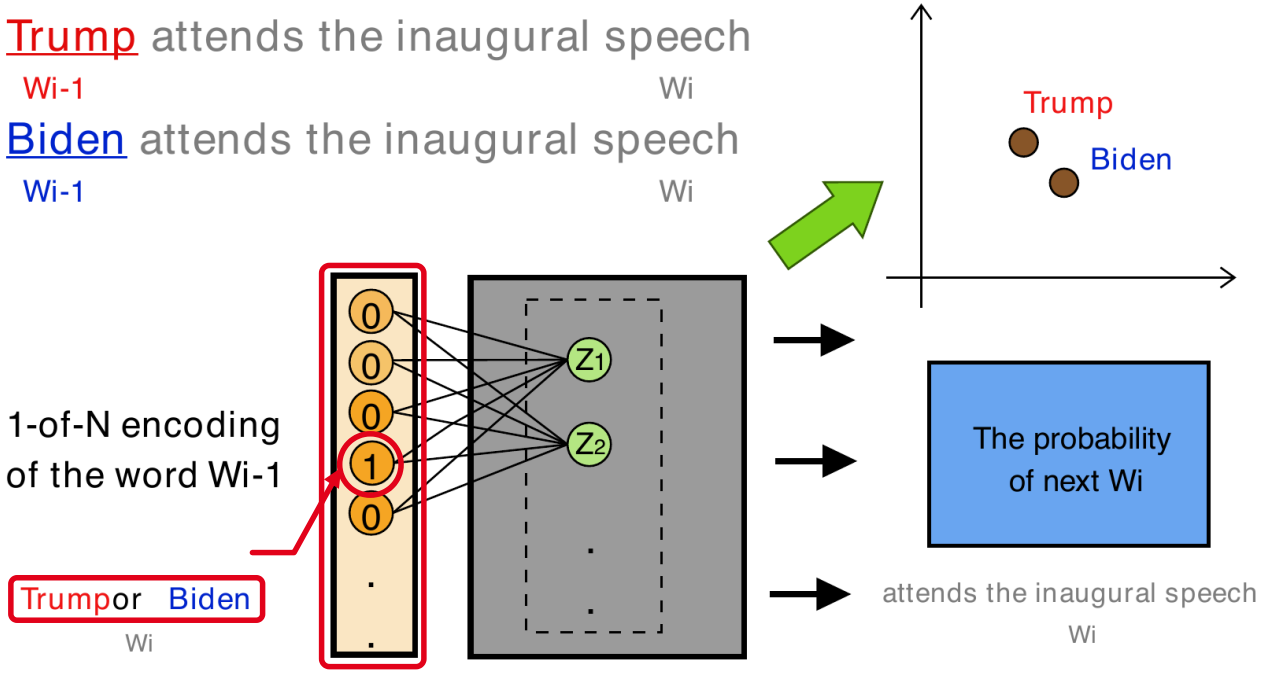
Rodman (2020) · *Political Analysis* · Case 2 · DOI
A Timely Intervention: Tracking the Changing Meanings of Political Concepts with Word Vectors

W10 optional readings used in W9

Müller & Proksch (2024) · *BJPS* · Case 3 · DOI
Nostalgia in European Party Politics: A Text-Based Measurement Approach

Widmann & Wich (2023) · *Political Analysis* · W10 comparison · DOI
Creating and Comparing Dictionary, Word Embedding, and Transformer-Based Models to Measure Discrete Emotions in German Political Text

Learn coordinates by predicting words



Input One active vocabulary position.

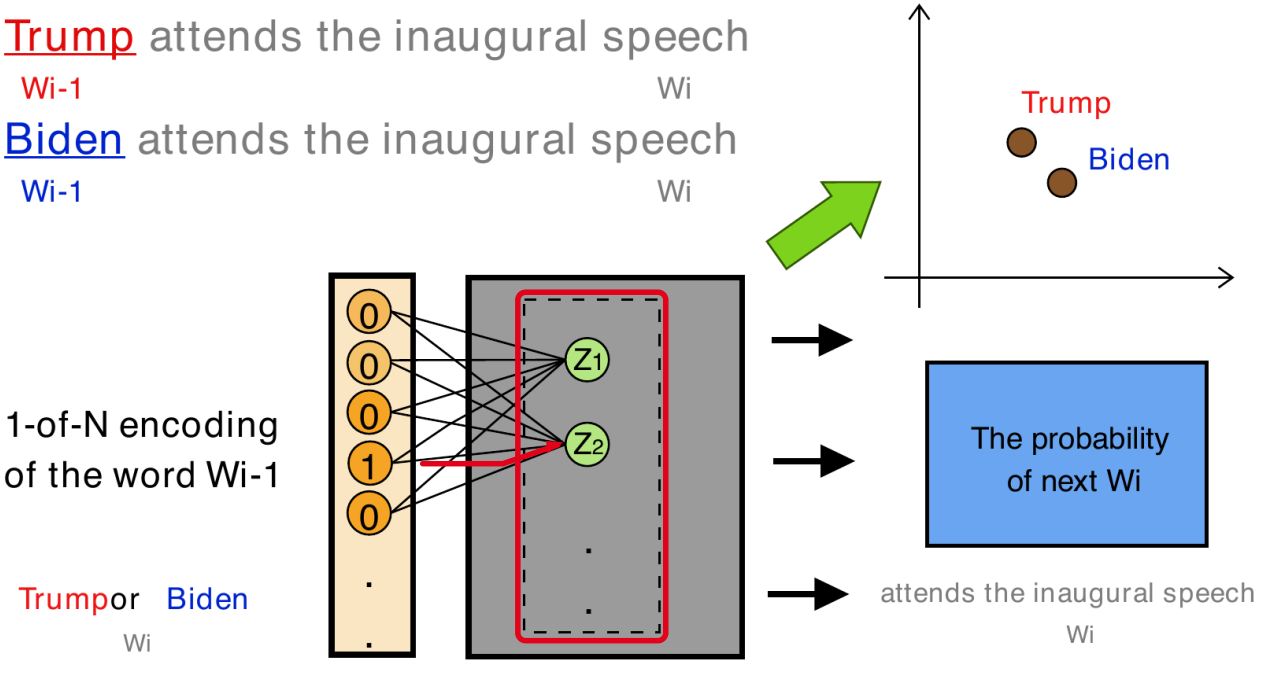
Hidden Learned coordinates z_1, z_2.

Output Next-word probabilities; target: *attends*.

Update Error updates the connection weights.

The plotted positions are a teaching illustration.

Learn coordinates by predicting words



Input One active vocabulary position.

Hidden Learned coordinates z_1, z_2.

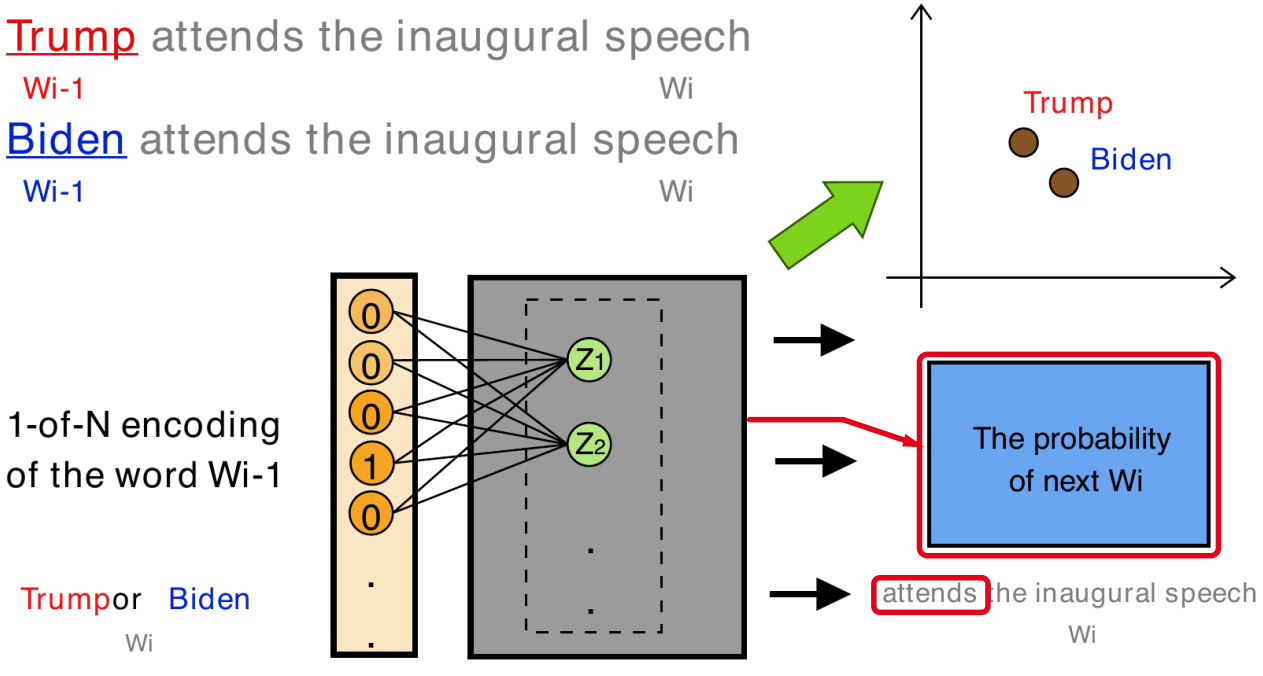
Output Next-word probabilities; target: *attends*.

Update Error updates the connection weights.

The plotted positions are a teaching illustration.

Flair token extraction

Learn coordinates by predicting words



Input One active vocabulary position.

Hidden Learned coordinates z_1, z_2.

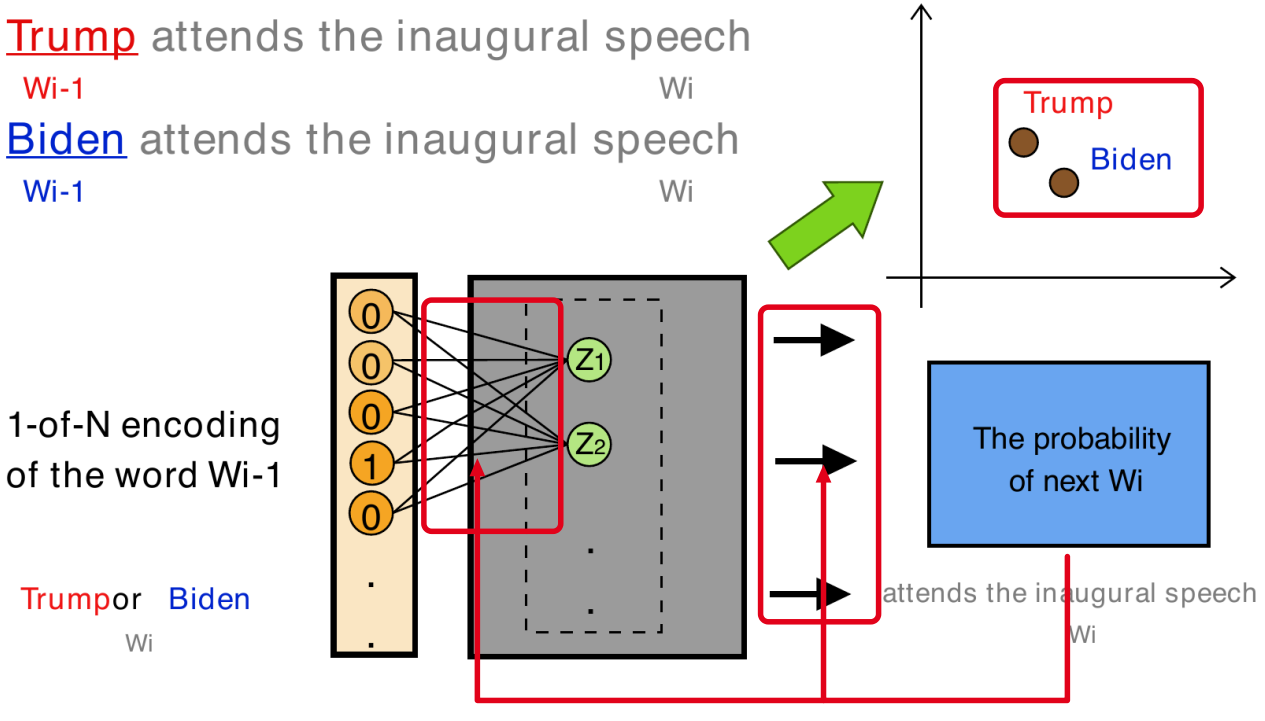
Output Next-word probabilities; target: *attends*.

Update Error updates the connection weights.

The plotted positions are a teaching illustration.

Flair token extraction

Learn coordinates by predicting words



Input One active vocabulary position.

Hidden Learned coordinates z_1, z_2.

Output Next-word probabilities; target: *attends*.

Update Error updates the connection weights.

The plotted positions are a teaching illustration.

Flair token extraction

Related words in prediction

The book's movie review example

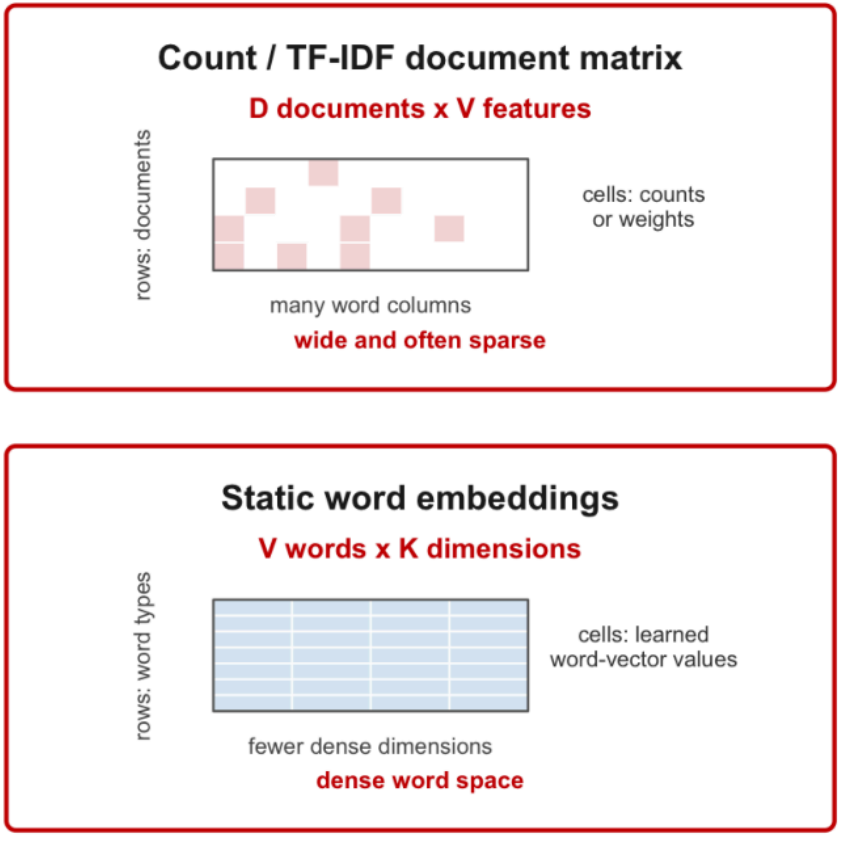
We train a classifier on labeled reviews that use **great**.

A new review uses **fantastic**. What can we carry over if the two words have related vectors?

We can predict using word similarities learned from the training corpus.

Grimmer, Roberts & Stewart (2022), §8.1, p. 80.

What does each row represent?



Document counts give each speech a row of feature counts.

Static word embeddings give each word type a row of learned coordinates.

Which rows do we compare for two speeches? For two words?

Schoonvelde (2026), *Comparing Representations*, ESS Day 3, original p. 6, document and word panels. ESS materials and page map

Compute cosine for two speeches

Speech	security	forces	energy
Defence	1	1	0
Energy	1	0	1

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\| \mathbf{a} \| \ \| \mathbf{b} \|} = \frac{1}{\sqrt{2}\sqrt{2}} = 0.50$$

Both speeches contain *security*. Each also contains a different feature.

Average word vectors to represent a document

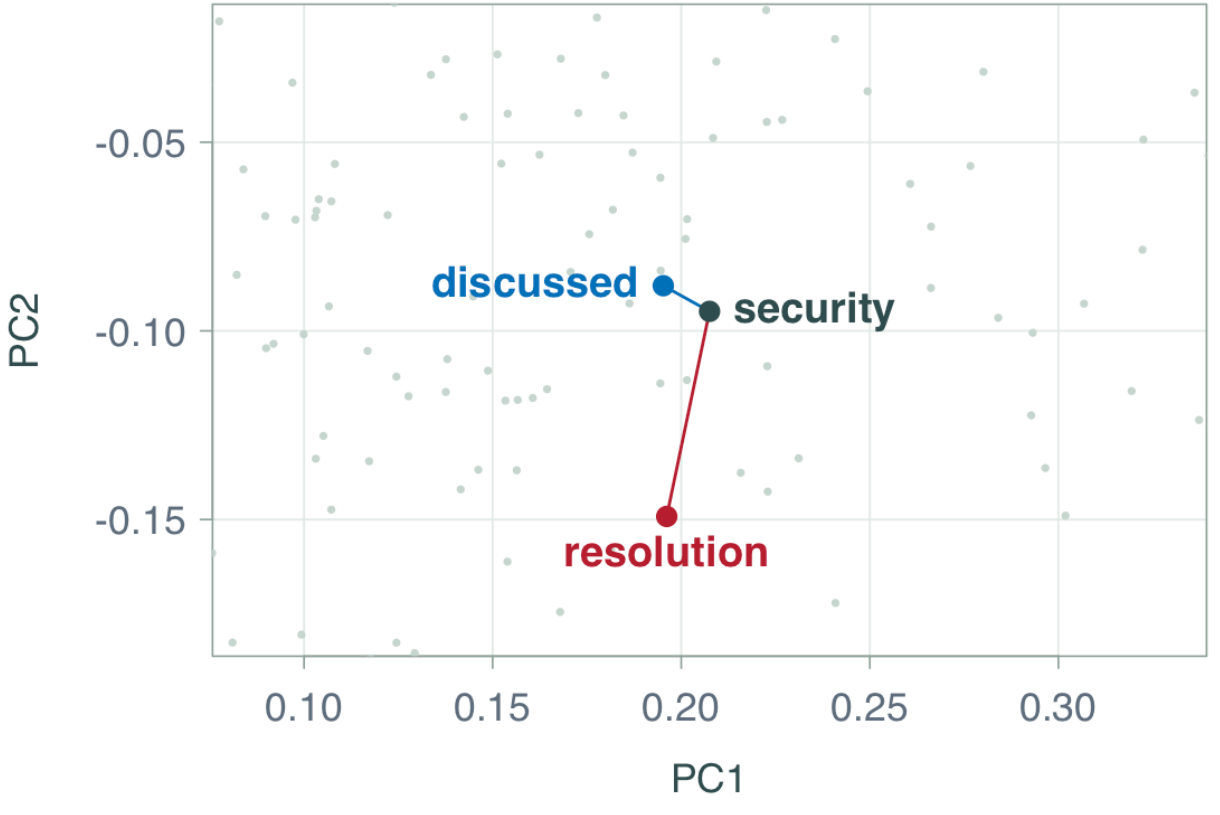
We average the available token vectors one coordinate at a time.

Token	Coordinate 1	Coordinate 2
first token	1	0
second token	0	2
document mean	**0.5**	**1.0**

Repeated tokens contribute repeatedly to the simple average. The resulting vector summarizes the document.

Grimmer, Roberts & Stewart (2022), §8.3, pp. 86–87. Two-coordinate arithmetic constructed for W9.

The PCA map changes the nearest words



On the **two-dimensional map**, *discussed* is closest to *security*.

Across all **30 dimensions**, *resolution* has the highest cosine with *security* (0.661).

PC1 and PC2 retain **9.54%** of the variance in this model's unit vectors.

How did presidents talk about manufacturing?

U.S. State of the Union addresses

The book compares **1900–1959** with **1960–2016**. It finds 53 mentions of *manufacturing* across the two periods.

1. Collect the words around each mention.
2. Represent those words with pretrained embeddings.
3. Average the context word vectors within each period and apply a learned linear transform to each average.

This à la carte procedure gives one **300-dimensional vector** for *manufacturing* per period.

Grimmer, Roberts & Stewart (2022), *Text as Data*, §8.2.7, p. 85

The neighbours of manufacturing change over time

Cosine distance ranks other words by their proximity to each period's *manufacturing* vector.

1900–1959	1960–2016
`exportation`	`retooling`
`oil-related`	`jobs`
`energy-intensive`	`offshoring`

The selected neighbours associate the term with trade and energy in the earlier period and restructuring and employment in the later one.

Selected reported neighbours from Grimmer, Roberts & Stewart (2022), *Text as Data*, §8.2.7, p. 85 and p. 86.

Validate the representation for its use

Evaluation	Question
Intrinsic	Do vector relationships agree with an external judgment about words?
Extrinsic	Do the vectors help on a downstream task, such as predicting labels?
Substantive	Does the comparison support the interpretation required by the research question?

For *manufacturing*, which passages could we read to check the period contrast?

Grimmer, Roberts & Stewart (2022), §8.4, p. 87.

W10 comparison: emotion measurement

W10 · Optional comparison

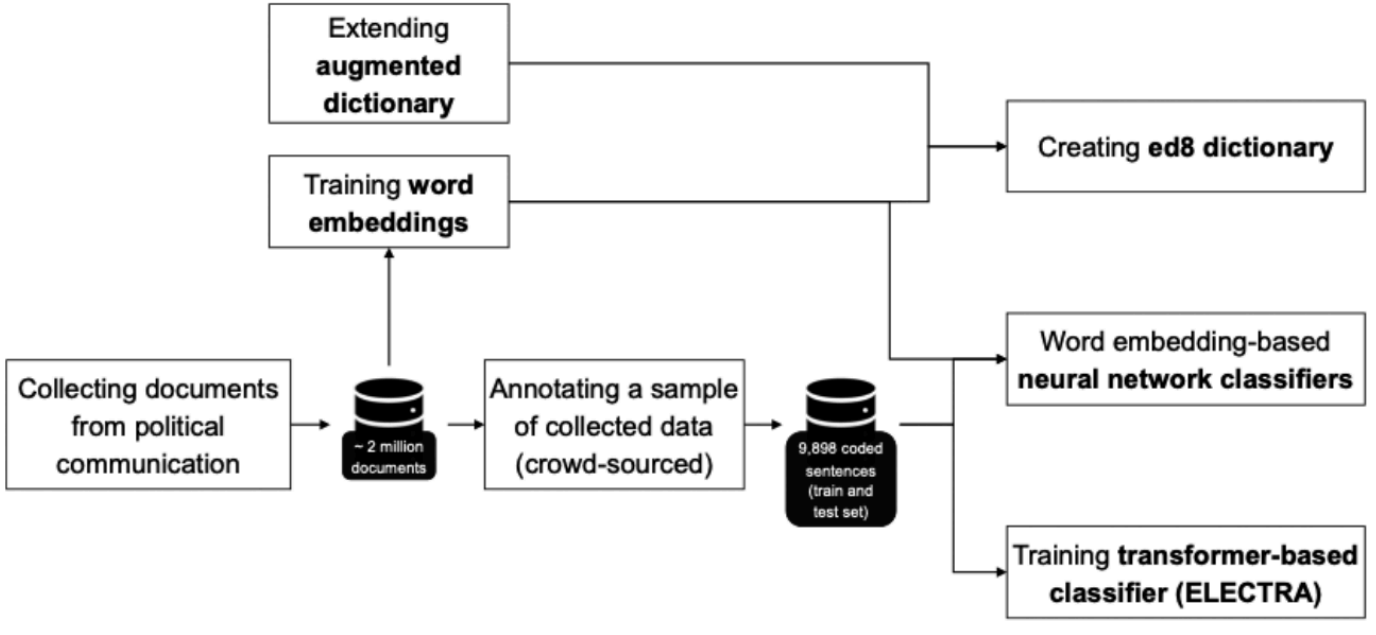


Figure 1. Overview over the three different approaches.

Question Which models distinguish eight emotional appeals in German political text?

Data 9,898 annotated sentences from parliamentary speeches and party Facebook posts.

Routes Neighbours expand ed8; word2vec sentence averages feed a neural network. ELECTRA is fine-tuned on annotated sentences.

W10 optional comparison reading · Widmann & Wich (2023), Figure 1, pp. 629–632 · Journal article heading

Test predictions against human coding

W10 · Optional comparison

Table 1. Precision, recall, and F1 scores for the three different approaches.

Emotions	Actual	Predicted	Precision	Recall	F1
ed8 dictionary					
Anger	508	281	0.83	0.46	0.59
Fear	189	287	0.43	0.66	0.52
Disgust	86	182	0.30	0.63	0.40
Sadness	201	289	0.41	0.59	0.48
Joy	143	179	0.46	0.58	0.52
Enthusiasm	220	248	0.44	0.50	0.47
Pride	158	247	0.31	0.48	0.38
Hope	305	303	0.53	0.53	0.53
Word-embeddings-based neural network approach					
Anger	508	500	0.80	0.78	0.79
Fear	189	152	0.61	0.49	0.55
Disgust	86	67	0.60	0.47	0.52
Sadness	201	122	0.70	0.42	0.53
Joy	143	92	0.68	0.44	0.54
Enthusiasm	220	176	0.64	0.51	0.57
Pride	158	123	0.52	0.41	0.46
Hope	305	265	0.69	0.60	0.64
Transformer-based (ELECTRA) approach					
Anger	508	495	0.85	0.83	0.84
Fear	189	221	0.60	0.70	0.64
Disgust	86	89	0.61	0.63	0.62
Sadness	201	181	0.64	0.57	0.60
Joy	143	122	0.70	0.59	0.64
Enthusiasm	220	242	0.62	0.68	0.65
Pride	158	151	0.61	0.58	0.60
Hope	305	352	0.68	0.78	0.73

F1 in the held-out sentences

Model	Anger	Fear
ed8	.59	.52
word2vec + NN	.79	.55
ELECTRA	.84	.64

The authors also tested a new random sample of sentences.

Widmann & Wich (2023), Table 1, p. 634. Robustness checks on pp. 635–638.

Four ways embeddings support an answer

Research goal	What gets a vector	How the study answers
1 · Locate parties	A party in a legislative term	PC1 gives a speech scale. Voteview helps interpret it.
2 · Track meaning	A word in a historical period	Cosine tracks associations. News passages show what changed.
3 · Measure nostalgia	A seed term or candidate term	Neighbours expand the dictionary. Its scores enter a regression.
4 · Compare word use	One use of a target word	Regression estimates the gap. Neighbours and passages explain it.

Rheault & Cochrane (2020) · Rodman (2020) · Rodriguez et al. (2023) · Müller & Proksch (2024)

Evaluate embeddings for the task

Use of the vectors	What to check
Features for prediction	Performance on held-out texts or a new corpus
Measures of political language	Human judgments, original passages and appropriate benchmarks
Comparisons across corpora	Domain, vocabulary coverage and stability across model settings
Social associations	Target and attribute sets, source-corpus composition and sensitivity of the comparison

Rodriguez and Spirling compare window sizes, dimensions, training runs and human judgments of word associations.

Rodriguez & Spirling (2022), pp. 105–108, 112–113 · Text as Data, §8.4

Campaign language and political knowledge

Question. Are voters closer to expert placements when parties use simpler manifestos?

- Bischof and Senninger compare German and Austrian manifestos using **LIX**.
- They compare voters' left–right placements of parties with expert placements.
- Simpler manifestos are associated with smaller placement gaps.

Bischof & Senninger (2018), pp. 477–481 and 486–490

LIX and Flesch score text in opposite directions

Measure	What it uses	Higher score
LIX	Mean words per sentence + percentage of long words	More complex
Flesch Reading Ease	Words per sentence and syllables per word	Easier to read

A sentence split can change both scores while the policy proposal stays the same.

Bischof & Senninger (2018), p. 479 · quanteda readability reference